ST185 Pin Chart

94 connectors · 67 faces sourced · 27 still a numbered list

Red +12 V Black ground White signal Blue sensor return Orange sensor supply Yellow switch signal Green CAN Grey shield drain

Dashed = spare cavity Cavity colour is the wire's insulation colour. From MASTER-wirelist.csv.

Bussed plug = a Deutsch DT plug whose bussed cap ties every pin together: a splice you can unplug. Each sits where its name says.

Devices by harness

Each plug belongs to one harness (Dan 2026-11-01). Fuse boxes, relays, buses, studs and factory taps are hubs: several harnesses land on them.

HARNESS	DEVICES AND PLUGS
AB-cabin	OEM ambient air temp sensor A1 pin 1 (front, 6 in pigtail) (A1); OEM A/C amplifier (A15); OEM auto A/C amplifier A32 pin 15 SG (A32); accelerator pedal sensor (APS-PEDAL); +5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) (BUS-5V); Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) (BUS-GNDOUT-A); Sensor-ground bussed plug B at the ECU, to ECU plug B pin 22 (BUS-GNDOUT-B) (BUS-GNDOUT-B); IG2 wake diode (1N5822) (D-IG2-WAKE); DRB102 bulkhead, cabin side (DRB102-CABIN); Link G4X plug A (ECU-A); Link G4X plug C, CAN / comms port (generic round 6-pin plug, centre dash) (ECU-C); Fuel sender inline plug, harness side, trunk (IX-FUEL) (IX-FUEL-P); Front wheel-speed inline plug, VRC side (IX-WS-F) (IX-WS-F-S); Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R) (IX-WS-R-S); Cam pull-up 1.8 k to +8 V (R-CAM-PULLUP); 470R fuel level pull-up to +5V (R-FUELLVL); Lock-sensor tap 4.7 k from tacho (R-LOCK); POWER box NEG bar, GND stud, ring 1 of 3 (T-PWR-NEG-GND-1); POWER box NEG bar, GND stud, ring 3 of 3 (T-PWR-NEG-GND-3)
AB-engine	Boost Control Solenoid (BOOST-SOL); Cam Hall Sensor (RacerX) (CAM); Link CAN-Lambda (CAN-LAMBDA); 120 ohm CAN terminator plug, Lambda end (CAN-TERM-END); Charge-Pipe IAT Sensor (CHARGE-IAT); Coolant Pressure Sensor (CLNT-P); Coil 1 (COIL-1); Coil 2 (COIL-2); Coil 3 (COIL-3); Coil 4 (COIL-4); Crank VR Sensor (36-2) (CRANK); DRB102 bulkhead, engine side (DRB102-ENGINE); ECT Sensor (ECT); Bosch 74.5mm ETB (ETB); Flex Fuel Sensor (FLEX); Fuel Pressure Sensor (FUEL-P); IAT Sensor (Manifold) (IAT); Injector 1 (INJ-1); Injector 2 (INJ-2); Injector 3 (INJ-3); Injector 4 (INJ-4); Knock Sensor 1 (KNOCK-1); MAP Sensor (MAP); MR-S EHPS pump plug B (MRS-B); MR-S EHPS pump plug C (MRS-C); Oil Pressure Sensor (OIL-P); Oil Temp Sensor (OIL-T); Reverse switch (EWD back-up SW B1) (REV-SW); Starter solenoid S (50) terminal, 1-way plug (STARTER-TRIGGER); Turbo Speed Sensor (TURBO-SPD); Gearbox speed sensor (VSS, Toyota 83181-20040) (VSS)
CabinPower	Pop-up diode D1 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug) (D-R8-1); Pop-up diode D2 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug) (D-R8-2); DRL switch (DRL-SW); OEM R-Y lead from cowl splice I2, cut near retract control relay R8 pin 8: I2 side (behind the LH dash) (I2); LOGIC fuse box relay M8 (DRL (both headlamps)) (K-DRL); LOGIC fuse box relay M1 (low beam (both headlamps)) (K-HL); LOGIC fuse box relay M2 (high beam (both headlamps)) (K-HR); LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) (K-POD); Dash high-beam LED feed resistor 220 ohm 1 W (R-HB); OEM retract control relay R8 pin 12 (behind the LH dash), W-B ground lead (R8); LOGIC box Bus 1, feed stud (T-LOG-BUS1); LOGIC box Bus 2, feed stud (T-LOG-BUS2); POWER box AMI IN-1 fuse, output stud (T-PWR-AMI-IN-1); POWER box AMI IN-2 fuse, output stud (T-PWR-AMI-IN-2); POWER box AMI IN-4 fuse, output stud (T-PWR-AMI-IN-4); POWER box AMI IN-5 fuse, output stud (T-PWR-AMI-IN-5); POWER box AMI OUT-1 fuse, output stud (T-PWR-AMI-OUT-1); POWER box AMI OUT-2 fuse, output stud (T-PWR-AMI-OUT-2); POWER box AMI OUT-3 fuse, output stud (T-PWR-AMI-OUT-3); POWER box AMI OUT-4 fuse, output stud (T-PWR-AMI-OUT-4); POWER box AMI OUT-5 fuse, output stud (T-PWR-AMI-OUT-5); POWER box AMI OUT-6 fuse, output stud (T-PWR-AMI-OUT-6); POWER box NEG bar, GND stud, ring 2 of 3 (T-PWR-NEG-GND-2)

HARNESS	DEVICES AND PLUGS
EngineRoom-C	A/C compressor A4 pin 4 (A4); OEM A5, factory Y-R lead (A5); Alternator G2 (ALT-PLUG); LH front side marker lamp F5 pin 1 (corner lamp) (F5); RH front side marker lamp F6 pin 1 (corner lamp) (F6); LH front turn / clearance lamp F7 pin 3 (F7); RH front turn / clearance lamp F8 pin 3 (F8); Front wiper motor F9 pin 3 (cowl, driver side) (F9); Uprated condenser fan (+) (FAN-COND); Uprated condenser fan (GND) (FAN-COND-GND); Uprated radiator fan (+) (FAN-RAD); Uprated radiator fan (GND) (FAN-RAD-GND); Ground stud EA, RH front fender (EWD ground point EA) (GND-EA); Ground stud EB, LH front fender (EWD ground point EB) (GND-EB); Ground stud EB, LH front fender, strap lug (EWD ground point EB) (GND-EB-2); LH headlamp H1 pin 1 (H1); RH headlamp H2 pin 2 (H2); Horn LH (EWD H3) (HORN-LH); Horn RH (EWD H4) (HORN-RH); MR-S EHPS pump plug A (MRS-A); LH retract motor R2 pin 2 (R2); RH retract motor R3 pin 2 (R3); Engine block ground stud, lug 2 (Route 1 point 5) (T-ENG-BLOCK-2); Washer motor W1 pin 1 (RH front corner, low) (W1)
HeavyDC	Battery positive terminal post, trunk RH rear (Route 1 point 1) (BATT-POS); Trunk relay box (BATT-RELAY-BLOCK); Ground bolt IG, right kick (EWD ground point IG) (GND-IG); Engine-bay jump post (JUMP-POST); RADLOK (RADLOK-NEG-CAB); RADLOK (RADLOK-NEG-ENG); RADLOK + firewall (RADLOK-POS-CAB); RADLOK + engine (RADLOK-POS-ENG); Alternator B+ stud, rear RH of engine (Route 1 point 3) (T-ALT-B); Engine block ground stud, lug 1 (Route 1 point 5) (T-ENG-BLOCK); POWER box NEG bar, BAT- stud (T-PWR-NEG-BAT); POWER box NEG bar, GND ID stud (T-PWR-NEG-ID); POWER box NEG bar, GND IG stud (T-PWR-NEG-IG); POWER box NEG bar, RADLOK- stud (T-PWR-NEG-RADLOK); POWER box + bar, BAT+ IN stud (T-PWR-POS-BAT); POWER box + bar, RADLOK+ stud (T-PWR-POS-RADLOK); Starter B+ stud, its one lug (Route 1 point 2) (T-STARTER-B); Battery negative terminal post, trunk RH rear (Route 1 point 1) (T-TRUNK-NEG)
RearFuel	Fuel Level Sender (FUEL-LEVEL); Fuel Pump (FUEL-PUMP); Ground point BL, back panel centre (EWD ground point BL) (GND-BL); Fuel sender inline plug, sender side, trunk (IX-FUEL) (IX-FUEL-S)
Dash	OEM cluster plug C13 (B) pin 2, G-B wire (C13); OEM cluster plug C14 (C) pin 9, B-O wire (C14); OEM combination switch plug C16 (B) pin 5 (cruise control switch), W wire (C16); OEM cruise control clutch switch C17 pin 2, B wire (C17); ECUMaster CSB3 (CSB3); Charge LED (D-CHG); High beam LED (D-HB); Oil pressure LED (D-OIL); Park brake LED (D-PK); Turn LH LED (D-TL); Turn RH LED (D-TR); RealDash Pi 5 A inline fuse (F-PI); OEM cowl plug II1 pin 3 (instrument panel centre, cowl side), GR (II1); Optocoupler board, 8 channel (Dan's, generic EL814) (OPTO-8); Cruise resistor on the CSB3 (R-CRUISE); OEM dimmer rheostat R10 pin 1 (left of the steering wheel), green TAIL (R10); OEM stop lamp switch S9 pin 1 (pedal), G-W wire (S9); Cluster CAN board CANH + CANL pins (old meter spot) (T-CLUSTER-CAN); Cluster buck converter IN- pin (old meter spot) (T-CLUSTER-GND-LEAD); Centre cluster headlight-sense input (J8 pin 15, GPIO28, active-low) (T-CLUSTER-HL); Cluster buck converter IN+ pin (old meter spot) (T-CLUSTER-PWR); Pi USB-CAN adapter CANH + CANL pins (center dash) (T-REALDASH-CAN); Pi 12-to-5 V supply IN- pin (center dash) (T-REALDASH-GND-LEAD); Pi 12-to-5 V supply IN+ pin (center dash) (T-REALDASH-PWR)
AntiTheft	Alarm constant +12 V point (AL-12V), at the alarm spot (AL-12V); Shock sensor constant +12 V point (AL-12V-SHK), at the alarm spot (AL-12V-SHK); Alarm door trigger point (AL-DOOR), at the alarm spot (AL-DOOR); Alarm ground point (AL-GND), at the alarm spot (AL-GND); Alarm hood trigger point (AL-HOOD), at the alarm spot (AL-HOOD); Hood pin pull-wire, front end (AL-HOOD-BAY), RH front corner, spot TBD (AL-HOOD-BAY); Alarm ignition +12 V point (AL-IGN), at the alarm spot (AL-IGN); Alarm immobiliser output point (AL-IMMOB), at the alarm spot (AL-IMMOB); Siren output point (AL-SIREN), at the alarm spot (AL-SIREN); Siren + pull-wire, engine-bay end (AL-SIREN-BAY), centre firewall behind the engine (AL-SIREN-BAY); Siren ground pull-wire, engine-bay end (AL-SIREN-GND-BAY), centre firewall behind the engine (AL-SIREN-GND-BAY); OEM door courtesy switch line (R-W) at J/B No.1 connector 1B pin 11 (left kick panel); 0 V with any door open (T-AT-DOOR); Ring terminal at the alarm controller; chassis ground point chosen by Dan at install (Dan N43 b) (T-AT-GND)
WheelSpeed-Front	Front wheel-speed inline plug, sensor side (IX-WS-F) (IX-WS-F-P); VRC front IN plug (VRC-F-IN); VRC front OUT plug (VRC-F-OUT); ABS Wheel Speed FL (WSS-FL); ABS Wheel Speed FR (WSS-FR)
WheelSpeed-Rear	Rear wheel-speed inline plug, sensor side (IX-WS-R) (IX-WS-R-P); VRC rear IN plug (VRC-R-IN); VRC rear OUT plug (VRC-R-OUT); ABS Wheel Speed RL (WSS-RL); ABS Wheel Speed RR (WSS-RR)

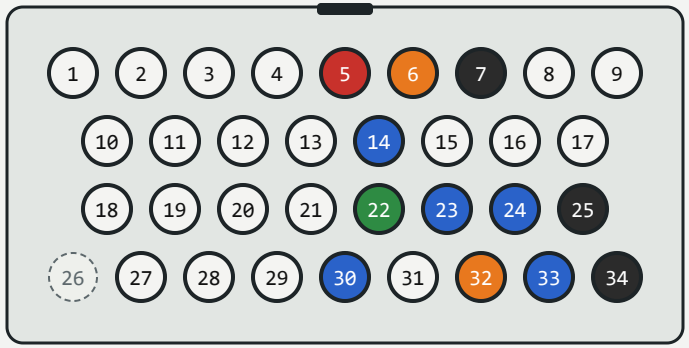
HARNESS	DEVICES AND PLUGS
Shared hubs	BUS-5V-2: AB-cabin, Dash; BUS-SW12: AB-cabin, CabinPower, RearFuel; ECU-B: AB-cabin, AntiTheft; GND-ID: Dash, HeavyDC; IE1: CabinPower, EngineRoom-C; IG2: AB-cabin, EngineRoom-C; JB1: AB-cabin, CabinPower, EngineRoom-C; K-ACC: AB-cabin, CabinPower, EngineRoom-C; K-EFI: AB-cabin, CabinPower; K-EPS: AB-cabin, CabinPower, EngineRoom-C; K-EPS-EN: AB-cabin, CabinPower; K-ETB: AB-cabin, CabinPower; K-FAN: AB-cabin, CabinPower, EngineRoom-C; K-FAN2: AB-cabin, CabinPower, EngineRoom-C; K-FP: AB-cabin, HeavyDC, RearFuel; K-HORN: CabinPower, EngineRoom-C; K-ON: AB-cabin, CabinPower; KILL-SW: AB-cabin, EngineRoom-C, HeavyDC; LOGIC: AB-cabin, CabinPower, Dash, EngineRoom-C; RB4: AB-cabin, CabinPower, EngineRoom-C; RLY-DRL: CabinPower, EngineRoom-C

AB-cabin

Link G4X plug A ECU-A · 4-1437290-1

FACE SOURCED

Link G4X plug A (centre dash)



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	w111_c	Injector 4 drive (-)	DRB102 bulkhead, cabin side · pin 24	3-1447221-4
2	20 White	w110_c	Injector 3 drive (-)	DRB102 bulkhead, cabin side · pin 23	3-1447221-4
3	20 White	w109_c	Injector 2 drive (-)	DRB102 bulkhead, cabin side · pin 22	3-1447221-4
4	20 White	w108_c	Injector 1 drive (-)	DRB102 bulkhead, cabin side · pin 21	3-1447221-4
5	18 Red	w_a5_pin	ECU main +12 V	Splice to: Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 2; DRB102 bulkhead, cabin side · pin 30; IG2 wake diode (1N5822) · pin 2	3-1447221-3
6	20 Orange	w19	+8 V sensor supply	Splice to: Cam pull-up 1.8 k to +8 V · pin 1; DRB102 bulkhead, cabin side · pin 63; DRB102 bulkhead, cabin side · pin 67	3-1447221-4
7	20 Black	w_drain_ecu_a	Shield drain	Splice to: Shield, not connected at the device end; Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 5; DRB102 bulkhead, cabin side · pin 76; DRB102 bulkhead, cabin side · pin 66; DRB102 bulkhead, cabin side · pin 57; DRB102 bulkhead, cabin side · pin 47	3-1447221-4
8	20 White	w200_c	Crank sensor signal	DRB102 bulkhead, cabin side · pin 74	3-1447221-4
9	20 White	w_trig2_pin	Cam sensor signal	Splice to: DRB102 bulkhead, cabin side · pin 64; Cam pull-up 1.8 k to +8 V · pin 2	3-1447221-4

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
10	20 White	w123_c	Coil 4 trigger (IGT)	DRB102 bulkhead, cabin side · pin 12	3-1447221-4
11	20 White	w122_c	Coil 3 trigger (IGT)	DRB102 bulkhead, cabin side · pin 11	3-1447221-4
12	20 White	w121_c	Coil 2 trigger (IGT)	DRB102 bulkhead, cabin side · pin 10	3-1447221-4
13	20 White	w120_c	Coil 1 trigger (IGT)	DRB102 bulkhead, cabin side · pin 9	3-1447221-4
14	20 Blue	w38	Pedal 1 signal main	accelerator pedal sensor · pin 6	3-1447221-4
15	20 White	w201_c	Coolant temp signal	DRB102 bulkhead, cabin side · pin 70	3-1447221-4
16	20 White	w202_c	Intake temp signal	DRB102 bulkhead, cabin side · pin 33	3-1447221-4
17	20 White	w26_c	MAP sensor signal	DRB102 bulkhead, cabin side · pin 17	3-1447221-4
18	20 White	w_ac_kill	A/C cancel (ACT)	OEM A/C amplifier	3-1447221-4
19	18 White	w_fp85	Fuel pump relay (-)	Trunk relay box relay K1 (fuel pump) · terminal 85	3-1447221-3
20	20 White	w81	ETB relay coil (-)	LOGIC fuse box relay M4 (ETB power) · terminal 85	3-1447221-4
21	20 White	w103_c	Boost solenoid (-)	DRB102 bulkhead, cabin side · pin 58	3-1447221-4
22	20 Green	w128_c	Throttle pos 1 main	DRB102 bulkhead, cabin side · pin 42	3-1447221-4
23	20 Blue	cab_fspur_l	FL conditioned out	Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 3	3-1447221-4
24	18 Blue	w68	Sensor ground	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 1	3-1447221-3
25	18 Black	w70	ECU power ground	Splice to: Link G4X plug A · pin 34; Link G4X plug B · pin 25; Link G4X plug B · pin 34; POWER box NEG bar, GND stud, ring 1 of 3	3-1447221-3
26	*spare*				

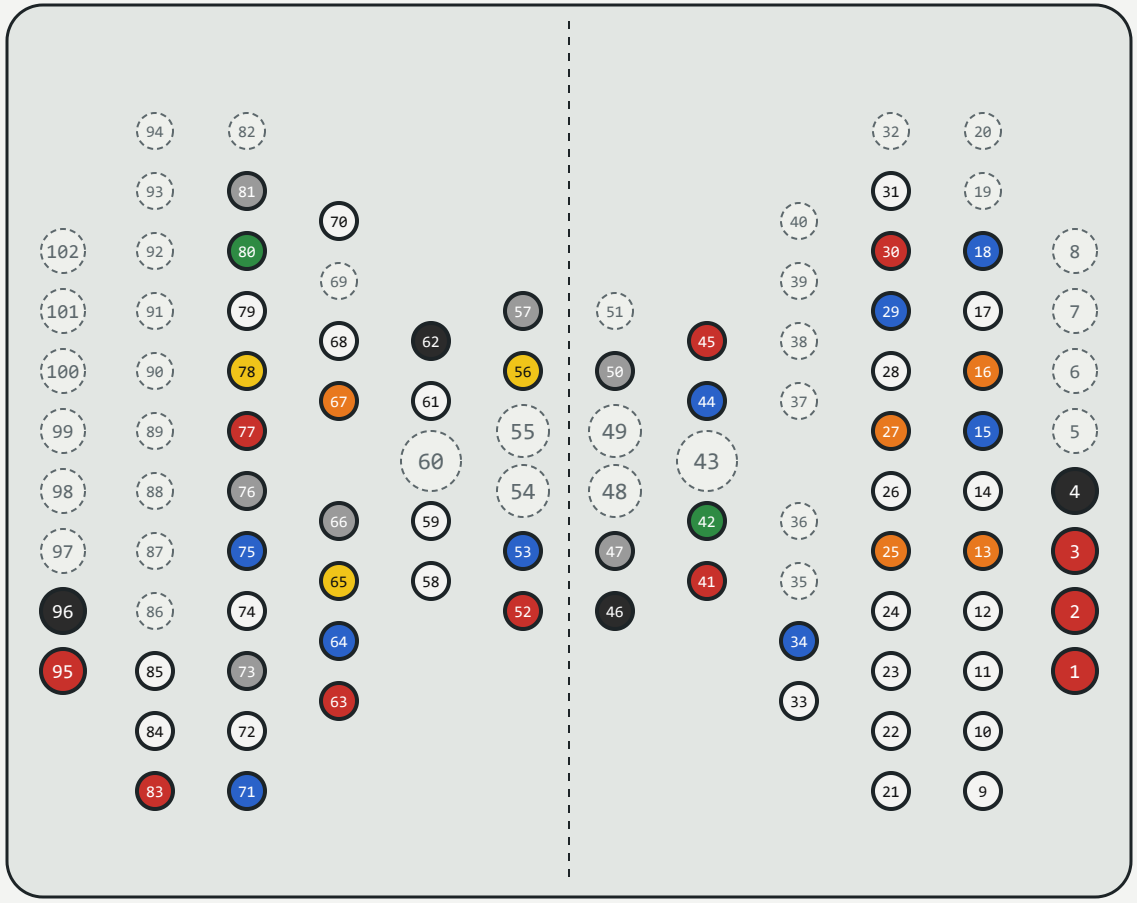
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
27	20 White	w_mrs_c	EHPS pump speed	DRB102 bulkhead, cabin side · pin 84	3-1447221-4
28	18 White	w76	EFI relay coil (-)	POWER fuse box relay R2 (EFI main) · terminal 85	3-1447221-3
29	18 White	w_fan85	Rad fan relay (-)	POWER fuse box relay R4 (radiator fan) · terminal 85	3-1447221-3
30	20 Blue	w16_c	Turbo speed signal	DRB102 bulkhead, cabin side · pin 53	3-1447221-4
31	20 White	w48_c	Flex fuel signal	DRB102 bulkhead, cabin side · pin 31	3-1447221-4
32	18 Orange	w67	+5 V sensor supply	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 1	3-1447221-3
33	20 Blue	w129_c	Throttle pos 2 sub	DRB102 bulkhead, cabin side · pin 44	3-1447221-4
34	18 Black	w_a34	ECU power ground	Splice to: Link G4X plug A · pin 25; Link G4X plug B · pin 25; Link G4X plug B · pin 34; POWER box NEG bar, GND stud, ring 1 of 3	3-1447221-3

Pin numbers: Link G4X XtremeX QSG p.5 pin numbering (1-34) · harness AB-cabin

DRB102 bulkhead, cabin side DRB102-CABIN · DRB12-102PAE-L018 + DRBF-1A

FACE SOURCED

DRB102 receptacle (RH firewall)



TE drawing view, numbers moulded on the insert

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	14 Red	w_cop_pwr_ac	Coil +12 V feed	LOGIC fuse box · fuse BUS2-3	0460-220-1231
2	14 Red	w_inj_pwr_ac	Injector +12 V feed	LOGIC fuse box · fuse BUS2-4	0460-220-1231
3	12 Red	w_strl_c	Starter solenoid S	OEM R/B No.4 starter relay pin 4 (RH kick), starter out	0460-220-1231

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
4	14 Black	w_cop_gnd_ac	Coil ground to the POWER box NEG bar	Splice to: DRB102 bulkhead, cabin side · pin 96; DRB102 bulkhead, cabin side · pin 62; POWER box NEG bar, GND stud, ring 3 of 3; ECUMaster CSB3 · pin 2; OEM combination switch plug C16 (B) pin 17 (cruise control switch), BR wire; Optocoupler board, 8 channel (Dan's, generic EL814) · pin COM1; Cluster buck converter IN- pin (old meter spot); Pi 12-to-5 V supply IN- pin (center dash); OEM A/C amplifier A15	0460-220-1231
5	*spare*				
6	*spare*				
7	*spare*				
8	*spare*				
9	20 White	w120_c	Coil 1 trigger (IGT)	Link G4X plug A · pin 13	0460-202-1631
10	20 White	w121_c	Coil 2 trigger (IGT)	Link G4X plug A · pin 12	0460-202-1631
11	20 White	w122_c	Coil 3 trigger (IGT)	Link G4X plug A · pin 11	0460-202-1631
12	20 White	w123_c	Coil 4 trigger (IGT)	Link G4X plug A · pin 10	0460-202-1631
13	20 Orange	w_5v_brail_c46	+5 V fuel pressure	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 6	0460-202-1631
14	20 White	w32_c	Fuel pressure signal	Link G4X plug B · pin 16	0460-202-1631
15	20 Blue	w_gnd_b_c21	Gnd to fuel pressure, gearbox speed	Sensor-ground bussed plug B at the ECU, to ECU plug B pin 22 (BUS-GNDOUT-B) · pin 8	0460-202-1631
16	20 Orange	w_5v_a_c39	+5 V MAP sensor	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 3	0460-202-1631
17	20 White	w26_c	MAP sensor signal	Link G4X plug A · pin 17	0460-202-1631
18	20 Blue	w_gnd_a_c44	Gnd MAP sensor	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 6	0460-202-1631
19	*spare*				
20	*spare*				
21	20 White	w108_c	Injector 1 drive (-)	Link G4X plug A · pin 4	0460-202-1631

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
22	20 White	`w109_c`	Injector 2 drive (-)	Link G4X plug A · pin 3	`0460-202-1631`
23	20 White	`w110_c`	Injector 3 drive (-)	Link G4X plug A · pin 2	`0460-202-1631`
24	20 White	`w111_c`	Injector 4 drive (-)	Link G4X plug A · pin 1	`0460-202-1631`
25	20 Orange	`w_5v_brail_c47`	+5 V coolant press	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 7	`0460-202-1631`
26	20 White	`w35_c`	Coolant press signal	Link G4X plug B · pin 23	`0460-202-1631`
27	20 Orange	`w_5v_brail_c45`	+5 V oil pressure	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 5	`0460-202-1631`
28	20 White	`w29_c`	Oil pressure signal	Link G4X plug B · pin 15	`0460-202-1631`
29	20 Blue	`w_gnd_b_c19`	Gnd to oil pressure, coolant pressure, oil temp, charge temp	Sensor-ground bussed plug B at the ECU, to ECU plug B pin 22 (BUS-GNDOUT-B) · pin 7	`0460-202-1631`
30	18 Red	`w_rail_sp_sw12_c`	+12 V boost + flex	Splice to: Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 2; IG2 wake diode (1N5822) · pin 2; Link G4X plug A · pin 5	`0460-202-1631`
31	20 White	`w48_c`	Flex fuel signal	Link G4X plug A · pin 31	`0460-202-1631`
32	*spare*				
33	20 White	`w202_c`	Intake temp signal	Link G4X plug A · pin 16	`0460-202-1631`
34	20 Blue	`w_rail_sp_gndout_c`	Gnd to intake temp, throttle sensor, coolant temp	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 2	`0460-202-1631`
35	*spare*				
36	*spare*				
37	*spare*				
38	*spare*				
39	*spare*				
40	*spare*				

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
41	20 Red	`w_rail_sp_5v_c`	+5 V throttle sensor	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 2	`0460-202-1631`
42	20 Green	`w128_c`	Throttle pos 1 main	Link G4X plug A · pin 22	`0460-202-1631`
43	*spare*				
44	20 Blue	`w129_c`	Throttle pos 2 sub	Link G4X plug A · pin 33	`0460-202-1631`
45	18 Red	`w124_c`	ETB motor +	Link G4X plug B · pin 18	`0460-202-1631`
46	18 Black	`w215_c`	ETB motor -	Link G4X plug B · pin 26	`0460-202-1631`
47	Shield	`w_shc_etbs`	Throttle shield	Splice to: Link G4X plug A · pin 7; Shield, not connected at the device end; Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 5; DRB102 bulkhead, cabin side · pin 76; DRB102 bulkhead, cabin side · pin 66; DRB102 bulkhead, cabin side · pin 57	`0460-202-1631`
48	*spare*				
49	*spare*				
50	Shield	`w_shc_etbm`	ETB motor shield	Splice to: DRB102 bulkhead, cabin side · pin 73; Link G4X plug B · pin 17; Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R) · pin 5	`0460-202-1631`
51	*spare*				
52	20 Red	`w_5v_a_c40`	+5 V turbo speed	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 4	`0460-202-1631`
53	20 Blue	`w16_c`	Turbo speed signal	Link G4X plug A · pin 30	`0460-202-1631`
54	*spare*				
55	*spare*				
56	20 Yellow	`w_gnd_a_c43`	Gnd turbo speed	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 5	`0460-202-1631`
57	Shield	`w_shc_turbo`	Turbo speed shield	Splice to: Link G4X plug A · pin 7; Shield, not connected at the device end; Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 5; DRB102 bulkhead, cabin side · pin 76; DRB102 bulkhead, cabin side · pin 66; DRB102 bulkhead, cabin side · pin 47	`0460-202-1631`
58	20 White	`w103_c`	Boost solenoid (-)	Link G4X plug A · pin 21	`0460-202-1631`
59	20 White	`w204_c`	Charge temp signal	Link G4X plug B · pin 7	`0460-202-1631`

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
60	*spare*				
61	20 White	w203_c	Oil temp signal	Link G4X plug B · pin 6	0460-202-1631
62	20 Black	w_flex_ret_c	12 V sensor return (flex fuel)	Splice to: DRB102 bulkhead, cabin side · pin 96; DRB102 bulkhead, cabin side · pin 4; POWER box NEG bar, GND stud, ring 3 of 3; ECUMaster CSB3 · pin 2; OEM combination switch plug C16 (B) pin 17 (cruise control switch), BR wire; Optocoupler board, 8 channel (Dan's, generic EL814) · pin COM1; Cluster buck converter IN- pin (old meter spot); Pi 12-to-5 V supply IN- pin (center dash); OEM A/C amplifier A15	0460-202-1631
63	20 Red	w_rail_sp_8v_c	+8 V to cam sensor	Splice to: Cam pull-up 1.8 k to +8 V · pin 1; Link G4X plug A · pin 6; DRB102 bulkhead, cabin side · pin 67	0460-202-1631
64	20 Blue	w_cam_sig_c	Cam sensor signal	Splice to: Link G4X plug A · pin 9; Cam pull-up 1.8 k to +8 V · pin 2	0460-202-1631
65	20 Yellow	w_gnd_a_c42	Gnd cam sensor	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 4	0460-202-1631
66	Shield	w_shc_cam	Cam sensor shield	Splice to: Link G4X plug A · pin 7; Shield, not connected at the device end; Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 5; DRB102 bulkhead, cabin side · pin 76; DRB102 bulkhead, cabin side · pin 57; DRB102 bulkhead, cabin side · pin 47	0460-202-1631
67	20 Orange	w_rail_sp_8v_vss_c	+8 V to gearbox speed sensor	Splice to: Cam pull-up 1.8 k to +8 V · pin 1; Link G4X plug A · pin 6; DRB102 bulkhead, cabin side · pin 63	0460-202-1631
68	20 White	w_vss_sig_c	Gearbox speed signal	Link G4X plug B · pin 29	0460-202-1631
69	*spare*				
70	20 White	w201_c	Coolant temp signal	Link G4X plug A · pin 15	0460-202-1631
71	20 Blue	w205_c	Knock sensor signal	Link G4X plug B · pin 9	0460-202-1631
72	20 White	w12_c	Knock sensor (-)	Sensor-ground bussed plug B at the ECU, to ECU plug B pin 22 (BUS-GNDOUT-B) · pin 6	0460-202-1631
73	Shield	w_shc_knock1	Knock sensor shield	Splice to: DRB102 bulkhead, cabin side · pin 50; Link G4X plug B · pin 17; Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R) · pin 5	0460-202-1631
74	20 White	w200_c	Crank sensor signal	Link G4X plug A · pin 8	0460-202-1631
75	20 Blue	w_gnd_a_c41	Gnd crank sensor	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 3	0460-202-1631

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
76	Shield	w_shc_crank	Crank sensor shield	Splice to: Link G4X plug A · pin 7; Shield, not connected at the device end; Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 5; DRB102 bulkhead, cabin side · pin 66; DRB102 bulkhead, cabin side · pin 57; DRB102 bulkhead, cabin side · pin 47	0460-202-1631
77	16 Red	w_rev_feed_c	+12 V to reverse switch	OEM IG2 pin 4 (under the ECU, cowl side)	0460-202-1631
78	16 Yellow	w_rev_sw_c	Switched +12 V from reverse switch	Splice to: Optocoupler board, 8 channel (Dan's, generic EL814) · pin IN2; OEM IG2 pin 13 (under the ECU, cowl side)	0460-202-1631
79	20 White	wc_h_bh_bc	CAN H to CAN-Lambda	Splice to: Link G4X plug C, CAN / comms port (generic round 6-pin plug, centre dash) · pin 3; ECUMaster CSB3 · pin 5; Cluster CAN board CANH + CANL pins (old meter spot); Pi USB-CAN adapter CANH + CANL pins (center dash)	0460-202-1631
80	20 Green	wc_l_bh_bc	CAN L to CAN-Lambda	Splice to: Link G4X plug C, CAN / comms port (generic round 6-pin plug, centre dash) · pin 4; ECUMaster CSB3 · pin 6; Cluster CAN board CANH + CANL pins (old meter spot); Pi USB-CAN adapter CANH + CANL pins (center dash)	0460-202-1631
81	Shield	cab_canbh_bc_sh	CAN-Lambda shield	Splice to: Link G4X plug C, CAN / comms port (generic round 6-pin plug, centre dash) · pin 1; Shield, not connected at the device end	0460-202-1631
82	*spare*				
83	18 Red	w_fb_mrs_en	EHPS enable +12 V	LOGIC fuse box relay M6 (EPS enable) · terminal 87	0460-202-1631
84	20 White	w_mrs_c	EHPS pump speed	Link G4X plug A · pin 27	0460-202-1631
85	18 White	w_eps_rly	EHPS relay coil (-)	EPS pump relay (beside the POWER fuse box) · terminal 85	0460-202-1631
86	*spare*				
87	*spare*				
88	*spare*				
89	*spare*				
90	*spare*				
91	*spare*				
92	*spare*				
93	*spare*				
94	*spare*				
95	14 Red	w_lam12_cp	CAN-Lambda +12 V	LOGIC fuse box relay M7 (ON-only feed (CAN-Lambda)) · terminal 87	0460-220-1231

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
96	14 Black	w_lamg_ac	CAN-Lambda ground	Splice to: DRB102 bulkhead, cabin side · pin 62; DRB102 bulkhead, cabin side · pin 4; POWER box NEG bar, GND stud, ring 3 of 3; ECUMaster CSB3 · pin 2; OEM combination switch plug C16 (B) pin 17 (cruise control switch), BR wire; Optocoupler board, 8 channel (Dan's, generic EL814) · pin COM1; Cluster buck converter IN- pin (old meter spot); Pi 12-to-5 V supply IN- pin (center dash); OEM A/C amplifier A15	0460-220-1231
97	*spare*				
98	*spare*				
99	*spare*				
100	*spare*				
101	*spare*				
102	*spare*				

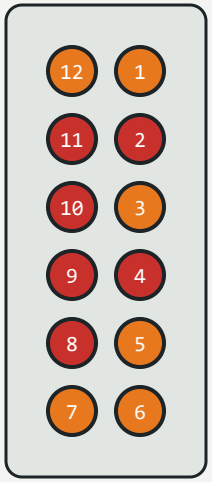
Pin numbers: TE drawing DRB12-102PXE-L018 rev C3 · harness AB-cabin

+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V)

`BUS-5V · DT06-12SA + W12S; DT04-12PA-P021 (bussed cap)`

FACE SOURCED

DT06-12 plug (ECU splice point)



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Orange	`w67`	+5 V from ECU A32	Link G4X plug A · pin 32	`0462-201-16141`
2	20 Red	`w_rail_sp_5v_c`	+5 V throttle sensor	DRB102 bulkhead, cabin side · pin 41	`0462-201-16141`
3	20 Orange	`w_5v_a_c39`	+5 V MAP sensor	DRB102 bulkhead, cabin side · pin 16	`0462-201-16141`
4	20 Red	`w_5v_a_c40`	+5 V turbo speed	DRB102 bulkhead, cabin side · pin 52	`0462-201-16141`
5	20 Orange	`w_5v_brail_c45`	+5 V oil pressure	DRB102 bulkhead, cabin side · pin 27	`0462-201-16141`
6	20 Orange	`w_5v_brail_c46`	+5 V fuel pressure	DRB102 bulkhead, cabin side · pin 13	`0462-201-16141`
7	20 Orange	`w_5v_brail_c47`	+5 V coolant press	DRB102 bulkhead, cabin side · pin 25	`0462-201-16141`
8	20 Red	`w_aps_vc1`	+5 V pedal sensor 1	accelerator pedal sensor · pin 4	`0462-201-16141`
9	20 Red	`w210`	+5 V pedal sensor 2	accelerator pedal sensor · pin 1	`0462-201-16141`
10	20 Red	`cab_fspur_5v`	+5 V front VRC	Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 1	`0462-201-16141`
11	20 Red	`cab_rspur_5v`	+5 V rear VRC	Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R) · pin 1	`0462-201-16141`
12	20 Orange	`w_5v_bus_link`	+5 V to BUS-5V-2	+5 V bussed plug 2 at the ECU, fed from BUS-5V pin 12 (BUS-5V-2) · pin 1	`0462-201-16141`

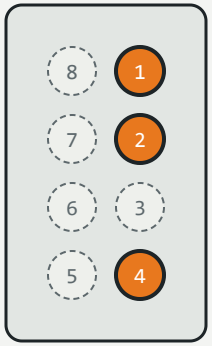
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-cabin

+5 V bussed plug 2 at the ECU, fed from BUS-5V pin 12 (BUS-5V-2)

`BUS-5V-2 · DT06-08SA + W8S; DT04-08PA-P021 (bussed cap)`

FACE SOURCED

DT06-8 plug (ECU splice point)



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Orange	`w_5v_bus_link`	+5 V from BUS-5V	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 12	`0462-201-16141`
2	20 Orange	`w_cru_5v_ac`	+5 V cruise ladder	Cruise resistor on the CSB3 · pin 1	`0462-201-16141`
4	20 Orange	`w44`	+5 V to fuel level pull-up	470R fuel level pull-up to +5V · pin 2	`0462-201-16141`
5	*spare*				
6	*spare*				
7	*spare*				
8	*spare*				

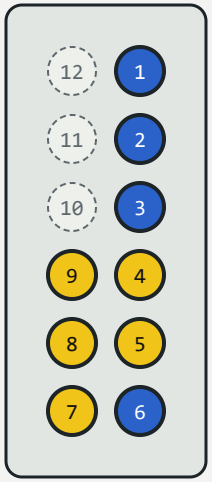
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-cabin, Dash

Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A)

`BUS-GNDOUT-A · DT06-12SA + W12S; DT04-12PA-P021 (bussed cap)`

FACE SOURCED

DT06-12 plug (ECU splice point)



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Blue	`w68`	Sensor gnd from A24	Link G4X plug A · pin 24	`0462-201-16141`
2	20 Blue	`w_rail_sp_gndout_c`	Gnd to intake temp, throttle sensor, coolant temp	DRB102 bulkhead, cabin side · pin 34	`0462-201-16141`
3	20 Blue	`w_gnd_a_c41`	Gnd crank sensor	DRB102 bulkhead, cabin side · pin 75	`0462-201-16141`
4	20 Yellow	`w_gnd_a_c42`	Gnd cam sensor	DRB102 bulkhead, cabin side · pin 65	`0462-201-16141`
5	20 Yellow	`w_gnd_a_c43`	Gnd turbo speed	DRB102 bulkhead, cabin side · pin 56	`0462-201-16141`
6	20 Blue	`w_gnd_a_c44`	Gnd MAP sensor	DRB102 bulkhead, cabin side · pin 18	`0462-201-16141`
7	20 Yellow	`w40`	Pedal 1 ground	accelerator pedal sensor · pin 5	`0462-201-16141`
8	20 Yellow	`w_aps_gnd2`	Pedal 2 ground	accelerator pedal sensor · pin 2	`0462-201-16141`
9	20 Yellow	`cab_fspur_gnd`	Front VRC ground	Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 2	`0462-201-16141`
10	*spare*				
11	*spare*				
12	*spare*				

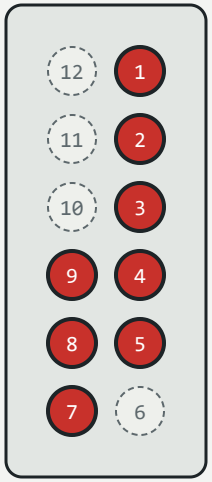
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-cabin

Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12)

`BUS-SW12 · DT06-12SA + W12S; DT04-12PA-P021 (bussed cap)`

FACE SOURCED

SW12 bus plug DT06-12 (glove box, by LOGIC box)



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	16 Red	`w_cp_sw12_in`	Switched +12 V feed	LOGIC fuse box · fuse BUS2-2	`0462-201-16141`
2	18 Red	`w_fb_sw12`	Switched +12 V feed	Splice to: IG2 wake diode (1N5822) · pin 2; Link G4X plug A · pin 5; DRB102 bulkhead, cabin side · pin 30	`0462-201-16141`
3	18 Red	`w_k_fp_86_cp`	Fuel pump relay coil +	Trunk relay box relay K1 (fuel pump) · terminal 86	`0462-201-16141`
4	18 Red	`w_k_fan_86_cp`	Radiator fan relay coil +	POWER fuse box relay R4 (radiator fan) · terminal 86	`0462-201-16141`
5	20 Red	`w_k_etb_86_cp`	ETB relay coil +	LOGIC fuse box relay M4 (ETB power) · terminal 86	`0462-201-16141`
7	18 Red	`w_k_fan2_86_cp`	Condenser fan relay coil +	POWER fuse box relay R5 (condenser fan) · terminal 86	`0462-201-16141`
8	18 Red	`w_k_eps_86_cp`	EHPS relay coil +	EPS pump relay (beside the POWER fuse box) · terminal 86	`0462-201-16141`
9	18 Red	`w_k_eps_en_86_cp`	EPS enable coil +	LOGIC fuse box relay M6 (EPS enable) · terminal 86	`0462-201-16141`

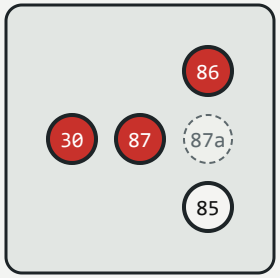
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
10	*spare*				
11	*spare*				
12	*spare*				

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-cabin, CabinPower, RearFuel

LOGIC fuse box relay M6 (EPS enable) K-EPS-EN · 6-1419137-4

FACE SOURCED

B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

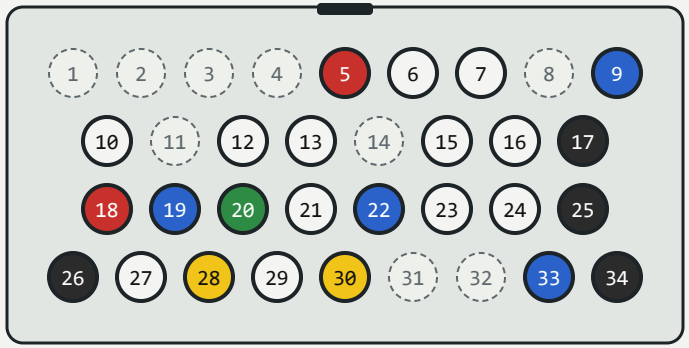
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	18 Red	w_fb_k_eps_en_30_cp	EPS enable +12 V in	LOGIC fuse box · fuse BUS2-7	42281-1
85	18 White	w_eps_trig	EPS enable relay (-)	Link G4X plug B · pin 12	280919-2
86	18 Red	w_k_eps_en_86_cp	EPS enable coil +	Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 9	280919-2
87	18 Red	w_fb_mrs_en	EHPS enable +12 V	DRB102 bulkhead, cabin side · pin 83	42281-1
87a	*spare*				

Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, CabinPower

Link G4X plug B ECU-B · 4-1437290-0

FACE SOURCED

ECU-B (Link G4X plug B, centre dash)



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	*spare*				
2	*spare*				
3	*spare*				
4	*spare*				
5	18 Red	w82_ac	V Ethrottle (ETB H-bridge supply)	LOGIC fuse box relay M4 (ETB power) · terminal 87	3-1447221-3
6	20 White	w203_c	Oil temp signal	DRB102 bulkhead, cabin side · pin 61	3-1447221-4
7	20 White	w204_c	Charge temp signal	DRB102 bulkhead, cabin side · pin 59	3-1447221-4
8	*spare*				
9	20 Blue	w205_c	Knock sensor signal	DRB102 bulkhead, cabin side · pin 71	3-1447221-4
10	20 White	w_tach	Engine speed (tacho)	Splice to: OEM A/C amplifier A15; Lock-sensor tap 4.7 k from tacho · pin 1	3-1447221-4
11	*spare*				
12	18 White	w_eps_trig	EPS enable relay (-)	LOGIC fuse box relay M6 (EPS enable) · terminal 85	3-1447221-3

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
13	18 White	w_fan285	Cond fan relay (-)	POWER fuse box relay R5 (condenser fan) · terminal 85	3-1447221-3
14	*spare*				
15	20 White	w29_c	Oil pressure signal	DRB102 bulkhead, cabin side · pin 28	3-1447221-4
16	20 White	w32_c	Fuel pressure signal	DRB102 bulkhead, cabin side · pin 14	3-1447221-4
17	20 Black	w_drain_ecu_b	Shield drain B	Splice to: DRB102 bulkhead, cabin side · pin 73; DRB102 bulkhead, cabin side · pin 50; Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R) · pin 5	3-1447221-4
18	18 Red	w124_c	ETB motor +	DRB102 bulkhead, cabin side · pin 45	3-1447221-3
19	20 Blue	cab_rspur_r	Rear right wheel spd	Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R) · pin 4	3-1447221-4
20	20 Green	cab_rspur_l	Rear left wheel spd	Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R) · pin 3	3-1447221-4
21	20 White	cab_fspur_r_ac	FR conditioned out	Front wheel-speed inline plug, VRC side (IX-WS-F) · pin 4	3-1447221-4
22	18 Blue	w69	Sensor ground B	Sensor-ground bussed plug B at the ECU, to ECU plug B pin 22 (BUS-GNDOUT-B) · pin 1	3-1447221-3
23	20 White	w35_c	Coolant press signal	DRB102 bulkhead, cabin side · pin 26	3-1447221-4
24	20 White	w_fl_sig_pin	Fuel level signal	Splice to: Fuel sender inline plug, harness side, trunk (IX-FUEL) · pin 1; 470R fuel level pull-up to +5V · pin 1	3-1447221-4
25	18 Black	w_b25_ac	ECU power ground	Splice to: Link G4X plug A · pin 25; Link G4X plug A · pin 34; Link G4X plug B · pin 34; POWER box NEG bar, GND stud, ring 1 of 3	3-1447221-3
26	18 Black	w215_c	ETB motor -	DRB102 bulkhead, cabin side · pin 46	3-1447221-3
27	20 White	w_at_immob2	Anti-theft input	Alarm immobiliser output point (AL-IMMOB), at the alarm spot	3-1447221-4
28	18 Yellow	w_ign	Ignition sw sense	OEM J/B No.1 (LH kick panel) 1H pin 7, B-O	3-1447221-3
29	20 White	w_vss_sig_c	Gearbox speed signal	DRB102 bulkhead, cabin side · pin 68	3-1447221-4
30	20 Yellow	w_start	Start request	LOGIC fuse box · fuse BUS2-13	3-1447221-4

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
31	*spare*				
32	*spare*				
33	20 Blue	`w_aps_vpa2`	Pedal 2 signal sub	accelerator pedal sensor · pin 3	`3-1447221-4`
34	18 Black	`w_b34_ac`	ECU power ground	Splice to: Link G4X plug A · pin 25; Link G4X plug A · pin 34; Link G4X plug B · pin 25; POWER box NEG bar, GND stud, ring 1 of 3	`3-1447221-3`

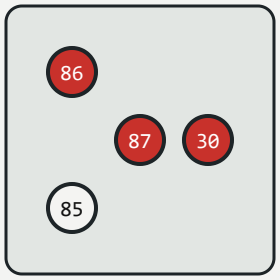
Pin numbers: Link G4X XtremeX QSG p.5 pin numbering (1-34) · harness AB-cabin, AntiTheft

Trunk relay box relay K1 (fuel pump)

K-FP · V23132-A2001-B200

FACE SOURCED

B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-suppressed so polarity-free



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 5 = 87).

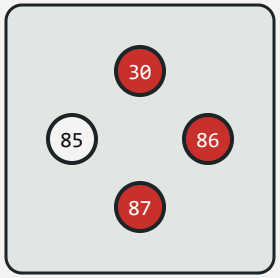
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	10 Red	w_fp_30	Fuel pump relay feed	Trunk relay box · FP fuse (40 A)	TBD
85	18 White	w_fp85	Fuel pump relay (-)	Link G4X plug A · pin 19	282110-1
86	18 Red	w_k_fp_86_cp	Fuel pump relay coil +	Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 3	282110-1
87	10 Red	w_fpump_r	Fuel pump +12 V	Fuel Pump · pin 1	TBD
87a	*spare*				

Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, HeavyDC, RearFuel

POWER fuse box relay R4 (radiator fan) K-FAN · 1-1414147-0

FACE SOURCED

B1-R4 Radiator fan relay (seated in the POWER box R4)



bottom (terminal side) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	10 Red	w_fan_pwr	Radiator fan +12 V	Uprated radiator fan (+) · pin 1	280756-4
85	18 White	w_fan85	Rad fan relay (-)	Link G4X plug A · pin 29	42281-1
86	18 Red	w_k_fan_86_cp	Radiator fan relay coil +	Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 4	42281-1
87	10 Red	w_fb_k_fan_87_cp	Radiator fan relay feed	POWER box AMI IN-4 fuse, output stud	280756-4

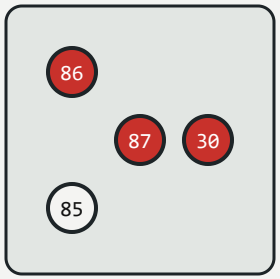
Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, CabinPower, EngineRoom-C

EPS pump relay (beside the POWER fuse box)

K-EPS · V23132-A2001-B200

FACE SOURCED

B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to ground by the MR-S pump pin B5 (R22); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole plug (TE terminals 1/2, not marked 85/86), resistor-suppressed so polarity-free



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 5 = 87).

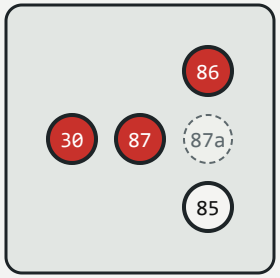
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	8 Red	w_fb_k_eps_30_ac	EHPS relay feed	POWER box AMI OUT-6 fuse, output stud	TBD
85	18 White	w_eps_rly	EHPS relay coil (-)	DRB102 bulkhead, cabin side · pin 85	282110-1
86	18 Red	w_k_eps_86_cp	EHPS relay coil +	Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 8	282110-1
87	8 Red	w_eps_pwr_bc	EHPS pump +12 V	MR-S EHPS pump plug A · pin 1	TBD
87a	*spare*				

Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, CabinPower, EngineRoom-C

LOGIC fuse box relay M5 (A/C clutch) K-ACC · 6-1419137-4

FACE SOURCED

B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B2-BUS2-8



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	16 Red	w_fb_k_acc_30_cp	+12 V to A/C clutch relay	Splice to: LOGIC fuse box relay M5 (A/C clutch) · terminal 86; LOGIC fuse box · fuse BUS2-8	42281-1
85	18 White	w_acc_trig	A/C clutch relay coil (-)	OEM A/C amplifier A15	280919-2
86	18 Red	w_acc86	A/C clutch relay coil+	Splice to: LOGIC fuse box relay M5 (A/C clutch) · terminal 30; LOGIC fuse box · fuse BUS2-8	280919-2
87	16 Red	w_acc_clutch	A/C clutch +12 V	A/C compressor A4 pin 4	42281-1
87a	*spare*				

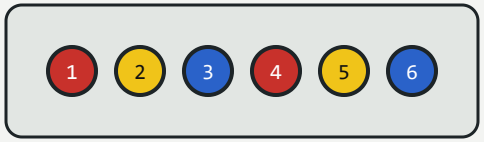
Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, CabinPower, EngineRoom-C

accelerator pedal sensor

APS-PEDAL · TBD

FACE NOT SOURCED

accelerator pedal sensor



layout not sourced: numbered list only, match the moulded numbers

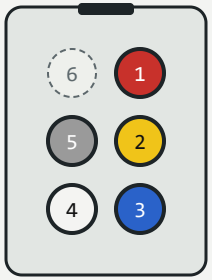
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	w210	+5 V (sensor 2)	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 9	TBD
2	20 Yellow	w_aps_gnd2	Gnd (sensor 2)	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 8	TBD
3	20 Blue	w_aps_vpa2	Signal 2 (sub)	Link G4X plug B · pin 33	TBD
4	20 Red	w_aps_vc1	+5 V (sensor 1)	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 8	TBD
5	20 Yellow	w40	Gnd (sensor 1)	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 7	TBD
6	20 Blue	w38	Signal 1 (main)	Link G4X plug A · pin 14	TBD

Pin numbers: TBD · harness AB-cabin

Front wheel-speed inline plug, VRC side (IX-WS-F) IX-WS-F-S · DT06-6S + W6S

FACE SOURCED

to front VR conditioner



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	cab_fspur_5v	VRC-F +5V	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 10	0462-201-1631
2	20 Yellow	cab_fspur_gnd	VRC-F GND	Sensor-ground bussed plug A at the ECU, to ECU plug A pin 24 (BUS-GNDOUT-A) · pin 9	0462-201-1631
3	20 Blue	cab_fspur_l	FL conditioned out	Link G4X plug A · pin 23	0462-201-1631
4	20 White	cab_fspur_r_ac	FR conditioned out	Link G4X plug B · pin 21	0462-201-1631
5	Shield	cab_fspur_sh	WS-F screen drain	Splice to: Link G4X plug A · pin 7; Shield, not connected at the device end; DRB102 bulkhead, cabin side · pin 76; DRB102 bulkhead, cabin side · pin 66; DRB102 bulkhead, cabin side · pin 57; DRB102 bulkhead, cabin side · pin 47	0462-201-1631
6	*spare*				

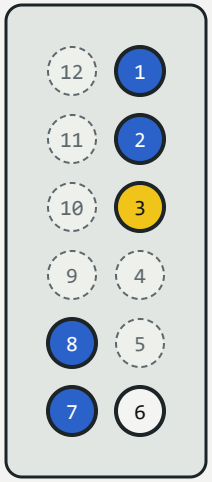
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-cabin

Sensor-ground bussed plug B at the ECU, to ECU plug B pin 22 (BUS-GNDOUT-B)

`BUS-GNDOUT-B · DT06-12SA + W12S; DT04-12PA-P021 (bussed cap)`

FACE SOURCED

DT06-12 plug (ECU splice point)



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Blue	`w69`	Sensor gnd from B22	Link G4X plug B · pin 22	`0462-201-16141`
2	20 Blue	`w45`	Fuel level ground	Fuel sender inline plug, harness side, trunk (IX-FUEL) · pin 2	`0462-201-16141`
3	20 Yellow	`cab_rspur_gnd`	Rear VRC ground	Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R) · pin 2	`0462-201-16141`
6	20 White	`w12_c`	Knock sensor (-)	DRB102 bulkhead, cabin side · pin 72	`0462-201-16141`
7	20 Blue	`w_gnd_b_c19`	Gnd to oil pressure, coolant pressure, oil temp, charge temp	DRB102 bulkhead, cabin side · pin 29	`0462-201-16141`
8	20 Blue	`w_gnd_b_c21`	Gnd to fuel pressure, gearbox speed	DRB102 bulkhead, cabin side · pin 15	`0462-201-16141`
9	*spare*				
10	*spare*				
11	*spare*				
12	*spare*				

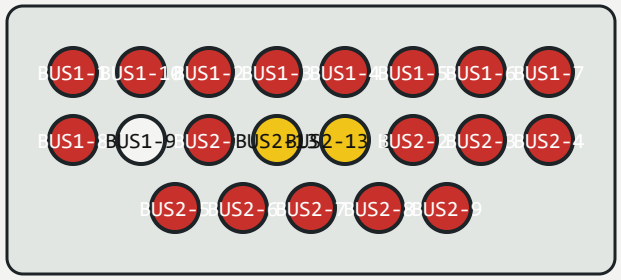
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-cabin

LOGIC fuse box

LOGIC · ST185-FB-LOGIC rev 7

FACE NOT SOURCED

B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 7 = models folder st185_fusebox_1101 revB, drawing ST185-FB-100L; DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
BUS1-1	18 Red	w_k_efi_86_cp	EFI relay coil +	POWER fuse box relay R2 (EFI main) · terminal 86	1-968849-1
BUS1-10	16 Red	w_fb_k_etb_30_cp	ETB relay +12 V in	LOGIC fuse box relay M4 (ETB power) · terminal 30	1-968851-1
BUS1-2	14 Red	w_hl_cp	Low beam +12 V feed	Splice to: LOGIC fuse box relay M1 (low beam (both headlamps)) · terminal 30; LOGIC fuse box relay M1 (low beam (both headlamps)) · terminal 86; DRL cut relay RLY-DRL (Dan N90 a) · pin 86	1-968851-1
BUS1-3	14 Red	w_hr_cp	High beam +12 V feed	Splice to: LOGIC fuse box relay M2 (high beam (both headlamps)) · terminal 30	1-968851-1
BUS1-4	14 Red	w_hz_cp	Hazard / horn +12 V feed	Splice to: OEM IE1 pin 12 (left kick, cowl side); LOGIC fuse box relay M3 (horn) · terminal 30; LOGIC fuse box relay M3 (horn) · terminal 86	1-968851-1
BUS1-5	14 Red	w_pmu_dome	Dome feed to J/B No.1	OEM J/B No.1 1H pin 2 (left kick)	1-968851-1
BUS1-6	14 Red	w_fb_k_on_30_cp	ON relay +12 V in	LOGIC fuse box relay M7 (ON-only feed (CAN-Lambda)) · terminal 30	1-968851-1
BUS1-7	20 Red	w_at_12v_tie_cp	Alarm +12 V	Splice to: Alarm constant +12 V point (AL-12V), at the alarm spot	1-968849-1
BUS1-8	20 Red	w_at_12v_shk_tie_cp	Shock sensor +12 V	Splice to: Shock sensor constant +12 V point (AL-12V-SHK), at the alarm spot	1-968849-1
BUS1-9	20 White	w_alt_s_c	Alt S battery sense	Alternator G2 · pin 1	1-968849-1

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
BUS2-1	14 Red	w_drl_cp	DRL +12 V feed	Splice to: LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 30; LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 86; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 86	1-968851-1
BUS2-13	20 Yellow	w_start	Start request	Link G4X plug B · pin 30	1-968849-1
BUS2-13 IN	20 Yellow	w_start_in_cp	Start sw ST1 feed	OEM ST1 lead B-W at old R/B No.4 pin 1, RH kick panel	1-968849-1
BUS2-2	16 Red	w_cp_sw12_in	Switched +12 V feed	Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 1	1-968851-1
BUS2-3	14 Red	w_cop_pwr_ac	Coil +12 V feed	DRB102 bulkhead, cabin side · pin 1	1-968851-1
BUS2-4	14 Red	w_inj_pwr_ac	Injector +12 V feed	DRB102 bulkhead, cabin side · pin 2	1-968851-1
BUS2-5	20 Red	w_csb12	CSB3 +12 V	Splice to: ECUMaster CSB3 · pin 1	1-968849-1
BUS2-6	16 Red	w_dev_clu	Cluster +12 V	Splice to: Cluster buck converter IN+ pin (old meter spot)	1-968851-1
BUS2-7	18 Red	w_fb_k_eps_en_30_cp	EPS enable +12 V in	LOGIC fuse box relay M6 (EPS enable) · terminal 30	1-968849-1
BUS2-8	16 Red	w_acc_fuse	A/C clutch relay +12 V feed (relay pins 30 + 86)	Splice to: LOGIC fuse box relay M5 (A/C clutch) · terminal 86; LOGIC fuse box relay M5 (A/C clutch) · terminal 30	1-968851-1
BUS2-9	20 Red	w_at_ig_tie_cp	Alarm ignition +12 V	Splice to: Alarm ignition +12 V point (AL-IGN), at the alarm spot	1-968849-1

Pin numbers: ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204) · harness AB-cabin, CabinPower, EngineRoom-C, Dash

Fuel sender inline plug, harness side, trunk (IX-FUEL)

`IX-FUEL-P · DT04-2P + W2P`

FACE SOURCED

IX-FUEL inline to fuel sender lead (trunk end)



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	`w_fl_sig`	Fuel level signal	Splice to: Link G4X plug B · pin 24; 470R fuel level pull-up to +5V · pin 1	`0460-202-1631`
2	20 Blue	`w45`	Fuel level ground	Sensor-ground bussed plug B at the ECU, to ECU plug B pin 22 (BUS-GNDOUT-B) · pin 2	`0460-202-1631`

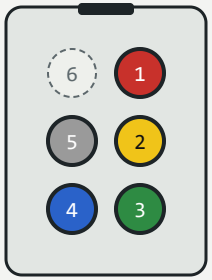
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-cabin

Rear wheel-speed inline plug, VRC side, dash above the ECU (IX-WS-R)

`IX-WS-R-S · DT06-6S + W6S`

FACE SOURCED

IX-WS-R inline to rear VRC (dash above ECU)



latch top

wire side (where the contacts go in) view · source

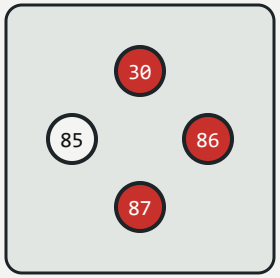
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	`cab_rspur_5v`	VRC-R +5V	+5 V bussed plug at the ECU, fed by ECU plug A pin 32 (BUS-5V) · pin 11	`0462-201-1631`
2	20 Yellow	`cab_rspur_gnd`	VRC-R GND	Sensor-ground bussed plug B at the ECU, to ECU plug B pin 22 (BUS-GNDOUT-B) · pin 3	`0462-201-1631`
3	20 Green	`cab_rspur_l`	Rear left wheel spd	Link G4X plug B · pin 20	`0462-201-1631`
4	20 Blue	`cab_rspur_r`	Rear right wheel spd	Link G4X plug B · pin 19	`0462-201-1631`
5	Shield	`cab_rspur_sh`	WS-R screen drain	Splice to: DRB102 bulkhead, cabin side · pin 73; DRB102 bulkhead, cabin side · pin 50; Link G4X plug B · pin 17	`0462-201-1631`
6	*spare*				

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-cabin

POWER fuse box relay R5 (condenser fan) K-FAN2 · 1-1393304-0

FACE SOURCED

B1-R5 Condenser fan relay (seated in the POWER box R5)



bottom (terminal side) view · source

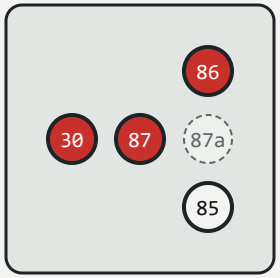
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	10 Red	w_fan2_pwr	Cond fan +12 V	Uprated condenser fan (+) · pin 1	280756-4
85	18 White	w_fan285	Cond fan relay (-)	Link G4X plug B · pin 13	42281-1
86	18 Red	w_k_fan2_86_cp	Condenser fan relay coil +	Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 7	42281-1
87	10 Red	w_fb_k_fan2_87_ac	Condenser fan relay feed	POWER box AMI IN-5 fuse, output stud	280756-4

Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, CabinPower, EngineRoom-C

LOGIC fuse box relay M4 (ETB power) K-ETB · 4-1904124-2

FACE SOURCED

B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

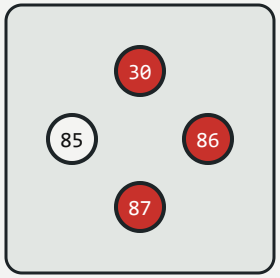
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	16 Red	w_fb_k_etb_30_cp	ETB relay +12 V in	LOGIC fuse box · fuse BUS1-10	42281-1
85	20 White	w81	ETB relay coil (-)	Link G4X plug A · pin 20	280919-2
86	20 Red	w_k_etb_86_cp	ETB relay coil +	Switched 12 V bussed plug in the glove box by the LOGIC fuse box, fed by LOGIC fuse BUS2-2 (BUS-SW12) · pin 5	280919-2
87	18 Red	w82_ac	V Ethrottle (ETB H-bridge supply)	Link G4X plug B · pin 5	42281-1
87a	*spare*				

Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, CabinPower

POWER fuse box relay R2 (EFI main) K-EFI · 1-1414147-0

FACE SOURCED

B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) permanent 12 V (LOGIC EFI86 5 A)



bottom (terminal side) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	10 Red	w77_cp	EFI relay k_efi (LOGIC Bus 2 feed)	LOGIC box Bus 2, feed stud	280756-4
85	18 White	w76	EFI relay coil (-)	Link G4X plug A · pin 28	42281-1
86	18 Red	w_k_efi_86_cp	EFI relay coil +	LOGIC fuse box · fuse BUS1-1	42281-1
87	12 Red	w_fb_k_efi_87_cp	EFI relay feed	POWER box AMI IN-2 fuse, output stud	280756-4

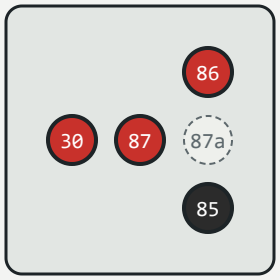
Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, CabinPower

LOGIC fuse box relay M7 (ON-only feed (CAN-Lambda))

K-ON · 6-1419137-4

FACE SOURCED

B2-M7 ON-only relay (seated in the LOGIC box M7) - coil+ (86) from J/B No.1 1H-7 IGN (live at ON and ST), coil- (85) to the POWER box NEG bar; feeds the CAN-Lambda



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	14 Red	w_fb_k_on_30_cp	ON relay +12 V in	LOGIC fuse box · fuse BUS1-6	42281-1
85	20 Black	w_k_on_85_cp	ON relay coil (-)	POWER box NEG bar, GND stud, ring 2 of 3	280919-2
86	18 Red	w_k_on_86_cp	IGN (ON / ST) +12 V	OEM J/B No.1 (LH kick panel) 1H pin 7, B-O	280919-2
87	14 Red	w_lam12_cp	CAN-Lambda +12 V	DRB102 bulkhead, cabin side · pin 95	42281-1
87a	*spare*				

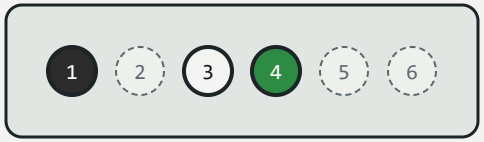
Pin numbers: ISO 7588 relay terminal numbers · harness AB-cabin, CabinPower

Link G4X plug C, CAN / comms port (generic round 6-pin plug, centre dash)

`ECU-C · none (generic round 6-pin plug, not bought)`

FACE NOT SOURCED

Link G4X plug C, CAN / comms port (generic round 6-pin plug, centre dash)



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Black	`w_shd_can_trunk`	CAN screen trunk	Splice to: Shield, not connected at the device end; DRB102 bulkhead, cabin side · pin 81	`none (contact comes with the plug)`
2	*spare*				
3	20 White	`wc_h_ecu`	CAN 1 high	Splice to: DRB102 bulkhead, cabin side · pin 79; ECUMaster CSB3 · pin 5; Cluster CAN board CANH + CANL pins (old meter spot); Pi USB-CAN adapter CANH + CANL pins (center dash)	`none (contact comes with the plug)`
4	20 Green	`wc_l_ecu`	CAN 1 low	Splice to: DRB102 bulkhead, cabin side · pin 80; ECUMaster CSB3 · pin 6; Cluster CAN board CANH + CANL pins (old meter spot); Pi USB-CAN adapter CANH + CANL pins (center dash)	`none (contact comes with the plug)`
5	*spare*				
6	*spare*				

Pin numbers: PCLink help page Communications Port Pinout (G4X / Xtreme / Fury wire-in), looking into the ECU connector · harness AB-cabin

Kill switch `KILL-SW · 52-45780`

FACE NOT SOURCED

Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1)



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
ALT1	16 Red	w_ig_ks_in	Alt IG to kill sw	OEM IG2 pin 15 (under the ECU, cowl side)	TBD
ALT2	16 Red	w_ig_ks_out	Alt IG excite	Alternator G2 · pin 3	TBD
MAIN1	1/0 Red	w_trunk_pos	battery +	Battery positive terminal post, trunk RH rear (Route 1 point 1)	TBD
MAIN2	10 Red	w_fp_feed	Battery relay box +	Trunk relay box · battery + stud	TBD
MAIN2	1/0 Red	w_ks_h1	battery +	POWER box + bar, BAT+ IN stud	TBD

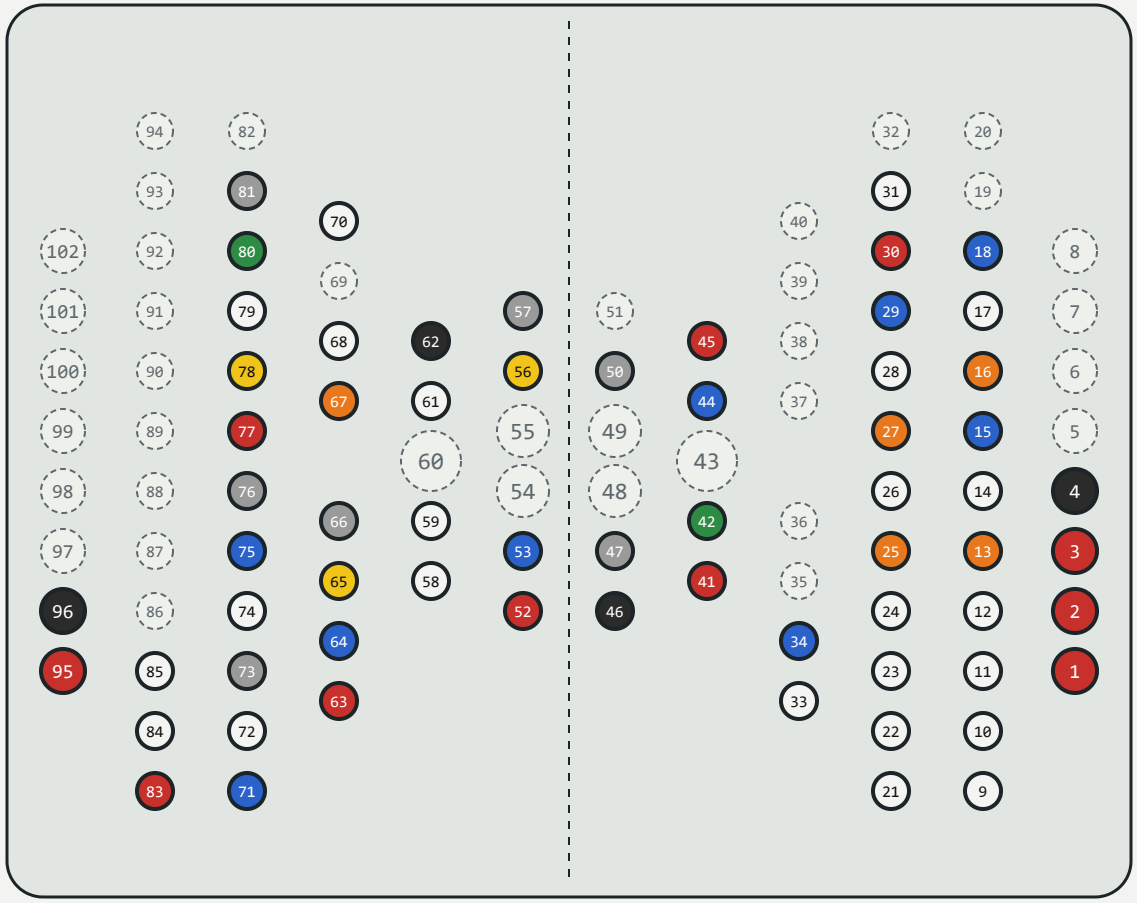
Pin numbers: TBD · harness AB-cabin, EngineRoom-C, HeavyDC

AB-engine

DRB102 bulkhead, engine side DRB102-ENGINE · DRB16-102SAE-L018

FACE SOURCED

BULKHEAD



TE drawing view, numbers moulded on the insert

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	14 Red	w_cop_pwr_e_ae	Coil +12 V feed	Splice to: Coil 1 · pin 1; Coil 2 · pin 1; Coil 3 · pin 1; Coil 4 · pin 1	0462-210-1231
2	14 Red	w_inj_pwr_e_ae	Injector +12 V feed	Splice to: Injector 1 · pin 1; Injector 2 · pin 1; Injector 3 · pin 1; Injector 4 · pin 1	0462-210-1231
3	12 Red	w_strl_e	Starter solenoid S	Starter solenoid S (50) terminal, 1-way plug · pin 1	0462-210-1231
4	14 Black	w_cop_gnd_e_ae	Coil ground to the POWER box NEG bar	Splice to: Coil 1 · pin 4; Coil 2 · pin 4; Coil 3 · pin 4; Coil 4 · pin 4	0462-210-1231

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
5	*spare*				
6	*spare*				
7	*spare*				
8	*spare*				
9	20 White	`w120_e`	Coil 1 trigger (IGT)	Coil 1 · pin 3	`0462-201-1631`
10	20 White	`w121_e`	Coil 2 trigger (IGT)	Coil 2 · pin 3	`0462-201-1631`
11	20 White	`w122_e`	Coil 3 trigger (IGT)	Coil 3 · pin 3	`0462-201-1631`
12	20 White	`w123_e`	Coil 4 trigger (IGT)	Coil 4 · pin 3	`0462-201-1631`
13	20 Orange	`w_5v_brail_e_c46`	+5 V fuel pressure	Fuel Pressure Sensor · pin 2	`0462-201-1631`
14	20 White	`w32_e`	Fuel pressure signal	Fuel Pressure Sensor · pin 3	`0462-201-1631`
15	20 Blue	`w_gnd_b_e_c21`	Gnd to fuel pressure, gearbox speed	Splice to: Fuel Pressure Sensor · pin 1; Gearbox speed sensor (VSS, Toyota 83181-20040) · pin 2	`0462-201-1631`
16	20 Orange	`w206`	+5 V MAP sensor	MAP Sensor · pin 2	`0462-201-1631`
17	20 White	`w26_e`	MAP sensor signal	MAP Sensor · pin 3	`0462-201-1631`
18	20 Blue	`w_gnd_a_e_c44`	Gnd MAP sensor	MAP Sensor · pin 1	`0462-201-1631`
19	*spare*				
20	*spare*				
21	20 White	`w108_e`	Injector 1 drive (-)	Injector 1 · pin 2	`0462-201-1631`
22	20 White	`w109_e`	Injector 2 drive (-)	Injector 2 · pin 2	`0462-201-1631`
23	20 White	`w110_e`	Injector 3 drive (-)	Injector 3 · pin 2	`0462-201-1631`
24	20 White	`w111_e`	Injector 4 drive (-)	Injector 4 · pin 2	`0462-201-1631`

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
25	20 Orange	`w_5v_brail_e_c47`	+5 V coolant press	Coolant Pressure Sensor · pin B	`0462-201-1631`
26	20 White	`w35_e`	Coolant press signal	Coolant Pressure Sensor · pin C	`0462-201-1631`
27	20 Orange	`w_5v_brail_e_c45`	+5 V oil pressure	Oil Pressure Sensor · pin 2	`0462-201-1631`
28	20 White	`w29_e`	Oil pressure signal	Oil Pressure Sensor · pin 3	`0462-201-1631`
29	20 Blue	`w_gnd_b_e_c19`	Gnd to oil pressure, coolant pressure, oil temp, charge temp	Splice to: Coolant Pressure Sensor · pin A; Oil Pressure Sensor · pin 1; Charge-Pipe IAT Sensor · pin 2; Oil Temp Sensor · pin 2	`0462-201-1631`
30	18 Red	`w_rail_sp_sw12_e`	+12 V boost + flex	Splice to: Boost Control Solenoid · pin 1; Flex Fuel Sensor · pin A	`0462-201-1631`
31	20 White	`w48_e`	Flex fuel signal	Flex Fuel Sensor · pin C	`0462-201-1631`
32	*spare*				
33	20 White	`w202_e`	Intake temp signal	IAT Sensor (Manifold) · pin 1	`0462-201-1631`
34	20 Blue	`w_rail_sp_gndout_e`	Gnd to intake temp, throttle sensor, coolant temp	Splice to: ECT Sensor · pin 2; Bosch 74.5mm ETB · pin 2; IAT Sensor (Manifold) · pin 2	`0462-201-1631`
35	*spare*				
36	*spare*				
37	*spare*				
38	*spare*				
39	*spare*				
40	*spare*				
41	20 Red	`w126`	+5 V throttle sensor	Bosch 74.5mm ETB · pin 3	`0462-201-1631`
42	20 Green	`w128_e`	Throttle pos 1 main	Bosch 74.5mm ETB · pin 6	`0462-201-1631`
43	*spare*				
44	20 Blue	`w129_e`	Throttle pos 2 sub	Bosch 74.5mm ETB · pin 5	`0462-201-1631`
45	18 Red	`w124_e_ae`	ETB motor +	Bosch 74.5mm ETB · pin 4	`0462-201-1631`
46	18 Black	`w215_e_ae`	ETB motor -	Bosch 74.5mm ETB · pin 1	`0462-201-1631`
47	Shield	`cab_etbs_sh`	Throttle shield	Shield, not connected at the device end	`0462-201-1631`

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
48	*spare*				
49	*spare*				
50	Shield	`cab_etbm_ae_sh`	ETB motor shield	Shield, not connected at the device end	`0462-201-1631`
51	*spare*				
52	20 Red	`w218`	+5 V turbo speed	Turbo Speed Sensor · pin A	`0462-201-1631`
53	20 Blue	`w16_e`	Turbo speed signal	Turbo Speed Sensor · pin B	`0462-201-1631`
54	*spare*				
55	*spare*				
56	20 Yellow	`w17`	Gnd turbo speed	Turbo Speed Sensor · pin C	`0462-201-1631`
57	Shield	`cab_turbo_sh`	Turbo speed shield	Shield, not connected at the device end	`0462-201-1631`
58	20 White	`w103_e`	Boost solenoid (-)	Boost Control Solenoid · pin 2	`0462-201-1631`
59	20 White	`w204_e`	Charge temp signal	Charge-Pipe IAT Sensor · pin 1	`0462-201-1631`
60	*spare*				
61	20 White	`w203_e`	Oil temp signal	Oil Temp Sensor · pin 1	`0462-201-1631`
62	20 Black	`w47`	12 V sensor return (flex fuel)	Flex Fuel Sensor · pin B	`0462-201-1631`
63	20 Red	`w_rail_sp_8v_e`	+8 V to cam sensor	Cam Hall Sensor (RacerX) · pin 1	`0462-201-1631`
64	20 Blue	`cab_cam_c2`	Cam sensor signal	Cam Hall Sensor (RacerX) · pin 2	`0462-201-1631`
65	20 Yellow	`w24`	Gnd cam sensor	Cam Hall Sensor (RacerX) · pin 3	`0462-201-1631`
66	Shield	`cab_cam_sh`	Cam sensor shield	Shield, not connected at the device end	`0462-201-1631`
67	20 Orange	`w_vss_8v`	+8 V to gearbox speed sensor	Gearbox speed sensor (VSS, Toyota 83181-20040) · pin 1	`0462-201-1631`
68	20 White	`w_vss_sig_e`	Gearbox speed signal	Gearbox speed sensor (VSS, Toyota 83181-20040) · pin 3	`0462-201-1631`
69	*spare*				
70	20 White	`w201_e`	Coolant temp signal	ECT Sensor · pin 1	`0462-201-1631`
71	20 Blue	`cab_knock1_c1`	Knock sensor signal	Knock Sensor 1 · pin 1	`0462-201-1631`

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
72	20 White	`cab_knock1_c2`	Knock sensor (-)	Knock Sensor 1 · pin 2	`0462-201-1631`
73	Shield	`cab_knock1_sh`	Knock sensor shield	Shield, not connected at the device end	`0462-201-1631`
74	20 White	`cab_crank_c1`	Crank sensor signal	Crank VR Sensor (36-2) · pin 1	`0462-201-1631`
75	20 Blue	`w2`	Gnd crank sensor	Crank VR Sensor (36-2) · pin 2	`0462-201-1631`
76	Shield	`cab_crank_sh`	Crank sensor shield	Shield, not connected at the device end	`0462-201-1631`
77	16 Red	`w_rev_feed_e`	+12 V to reverse switch	Reverse switch (EWD back-up SW B1) · pin 2	`0462-201-1631`
78	16 Yellow	`w_sw7_e`	Switched +12 V from reverse switch	Reverse switch (EWD back-up SW B1) · pin 1	`0462-201-1631`
79	20 White	`wc_h_lam_be`	CAN H to CAN-Lambda	Splice to: 120 ohm CAN terminator plug, Lambda end · pin 1; Link CAN-Lambda · pin 4	`0462-201-1631`
80	20 Green	`wc_l_lam_be`	CAN L to CAN-Lambda	Splice to: 120 ohm CAN terminator plug, Lambda end · pin 2; Link CAN-Lambda · pin 3	`0462-201-1631`
81	Shield	`cab_lamcan_be_sh`	CAN-Lambda shield	Shield, not connected at the device end	`0462-201-1631`
82	*spare*				
83	18 Red	`w_hv_mrs_en_e`	EHPS enable +12 V	MR-S EHPS pump plug C · pin 1	`0462-201-1631`
84	20 White	`w_mrs_e`	EHPS pump speed	MR-S EHPS pump plug B · pin 2	`0462-201-1631`
85	18 White	`w_eps_rly_e`	EHPS relay coil (-)	MR-S EHPS pump plug B · pin 5	`0462-201-1631`
86	*spare*				
87	*spare*				
88	*spare*				
89	*spare*				
90	*spare*				
91	*spare*				
92	*spare*				
93	*spare*				
94	*spare*				
95	14 Red	`w_lam12_be14`	CAN-Lambda +12 V	Splice to: Link CAN-Lambda · pin 1	`0462-210-1231`

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
96	14 Black	w_lamg_be14	CAN-Lambda ground	Splice to: Link CAN-Lambda · pin 2	0462-210-1231
97	*spare*				
98	*spare*				
99	*spare*				
100	*spare*				
101	*spare*				
102	*spare*				

Pin numbers: TE drawing DRB12-102PXE-L018 rev C3 · harness AB-engine

Crank VR Sensor (36-2)

CRANK · 90980-10947

FACE SOURCED

Crank VR Sensor (36-2)



latch top

mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	cab_crank_c1	Signal	DRB102 bulkhead, engine side · pin 74	in kit CONN-75800
2	20 Blue	w2	Gnd	DRB102 bulkhead, engine side · pin 75	in kit CONN-75800

Pin numbers: TBD · harness AB-engine

Cam Hall Sensor (RacerX) CAM · DT06-3S-CE05 + W3S

FACE SOURCED

Cam Hall Sensor (RacerX)



latch top

wire side (where the contacts go in) view. Deutsch moulds this 3-way as A / B / C; the master numbers it 1 / 2 / 3, read here as A = 1, B = 2, C = 3. · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	w_rail_sp_8v_e	+8 V	DRB102 bulkhead, engine side · pin 63	0462-201-16141
2	20 Blue	cab_cam_c2	Signal	DRB102 bulkhead, engine side · pin 64	0462-201-16141
3	20 Yellow	w24	Gnd	DRB102 bulkhead, engine side · pin 65	0462-201-16141

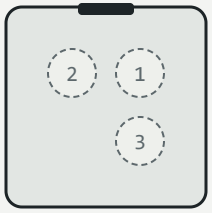
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-engine

Turbo Speed Sensor

TURBO-SPD · DT06-3S-CE05 + W3S

FACE SOURCED

Turbo Speed Sensor



latch top

wire side (where the contacts go in) view. Deutsch moulds this 3-way as A / B / C; the master numbers it 1 / 2 / 3, read here as A = 1, B = 2, C = 3. · source

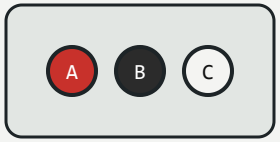
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
A	20 Red	w218	+5 V	DRB102 bulkhead, engine side · pin 52	0462-201-16141
B	20 Blue	w16_e	Signal	DRB102 bulkhead, engine side · pin 53	0462-201-16141
C	20 Yellow	w17	Gnd	DRB102 bulkhead, engine side · pin 56	0462-201-16141

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-engine

Flex Fuel Sensor FLEX · 13519047

FACE NOT SOURCED

Flex Fuel Sensor



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
A	20 Red	w217	+12 V	Splice to: DRB102 bulkhead, engine side · pin 30; Boost Control Solenoid · pin 1	15326426
B	20 Black	w47	12 V sensor return	DRB102 bulkhead, engine side · pin 62	15326426
C	20 White	w48_e	Signal	DRB102 bulkhead, engine side · pin 31	15326426

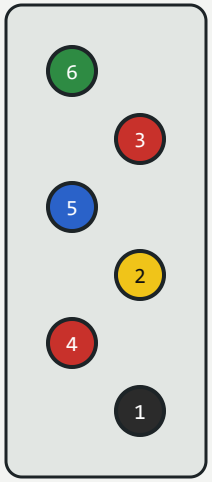
Pin numbers: device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Flex fuel pin table) · harness AB-engine

Bosch 74.5mm ETB

ETB · D 261 205 358-01

FACE SOURCED

Bosch 74.5mm ETB



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Black	w215_e_ae	Motor -	DRB102 bulkhead, engine side · pin 46	5-965906-1
2	20 Yellow	w127	Gnd	Splice to: DRB102 bulkhead, engine side · pin 34; ECT Sensor · pin 2; IAT Sensor (Manifold) · pin 2	in kit D 261 205 358-01
3	20 Red	w126	+5 V	DRB102 bulkhead, engine side · pin 41	in kit D 261 205 358-01
4	18 Red	w124_e_ae	Motor +	DRB102 bulkhead, engine side · pin 45	5-965906-1
5	20 Blue	w129_e	Position 2 (sub)	DRB102 bulkhead, engine side · pin 44	in kit D 261 205 358-01
6	20 Green	w128_e	Position 1 (main)	DRB102 bulkhead, engine side · pin 42	in kit D 261 205 358-01

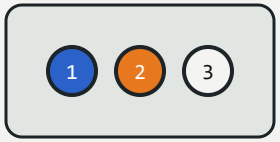
Pin numbers: Bosch Motorsport ETB data sheet 51048715 (pins 1A-6F) · harness AB-engine

MAP Sensor

MAP · 354404

FACE NOT SOURCED

MAP Sensor



layout not sourced: numbered list only, match the moulded numbers

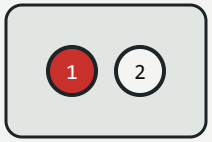
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	w_gnd_a_e_c44	Gnd MAP sensor	DRB102 bulkhead, engine side · pin 18	in kit 354404
2	20 Orange	w206	+5 V	DRB102 bulkhead, engine side · pin 16	in kit 354404
3	20 White	w26_e	Signal	DRB102 bulkhead, engine side · pin 17	in kit 354404

Pin numbers: device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (MAP pin table) · harness AB-engine

Injector 1 INJ-1 · CONN-76270

FACE NOT SOURCED

Injector 1



layout not sourced: numbered list only, match the moulded numbers

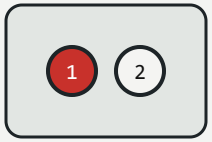
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Red	w211	+12 V	Splice to: DRB102 bulkhead, engine side · pin 2; Injector 2 · pin 1; Injector 3 · pin 1; Injector 4 · pin 1	in kit CONN-76270
2	20 White	w108_e	Drive (-)	DRB102 bulkhead, engine side · pin 21	in kit CONN-76270

Pin numbers: TBD · harness AB-engine

Injector 2 INJ-2 · CONN-76270

FACE NOT SOURCED

Injector 2



layout not sourced: numbered list only, match the moulded numbers

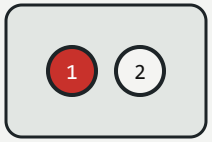
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Red	w212	+12 V	Splice to: DRB102 bulkhead, engine side · pin 2; Injector 1 · pin 1; Injector 3 · pin 1; Injector 4 · pin 1	in kit CONN-76270
2	20 White	w109_e	Drive (-)	DRB102 bulkhead, engine side · pin 22	in kit CONN-76270

Pin numbers: TBD · harness AB-engine

Injector 3 INJ-3 · CONN-76270

FACE NOT SOURCED

Injector 3



layout not sourced: numbered list only, match the moulded numbers

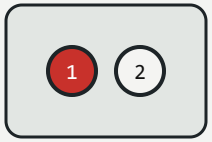
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Red	w213	+12 V	Splice to: DRB102 bulkhead, engine side · pin 2; Injector 1 · pin 1; Injector 2 · pin 1; Injector 4 · pin 1	in kit CONN-76270
2	20 White	w110_e	Drive (-)	DRB102 bulkhead, engine side · pin 23	in kit CONN-76270

Pin numbers: TBD · harness AB-engine

Injector 4 INJ-4 · CONN-76270

FACE NOT SOURCED

Injector 4



layout not sourced: numbered list only, match the moulded numbers

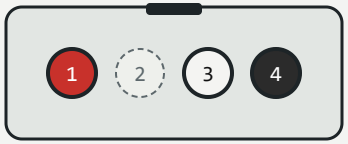
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Red	w214	+12 V	Splice to: DRB102 bulkhead, engine side · pin 2; Injector 1 · pin 1; Injector 2 · pin 1; Injector 3 · pin 1	in kit CONN-76270
2	20 White	w111_e	Drive (-)	DRB102 bulkhead, engine side · pin 24	in kit CONN-76270

Pin numbers: TBD · harness AB-engine

Coil 1 COIL-1 · 90980-11885

FACE SOURCED

COP 1 (1ZZ)



latch top

mating face view

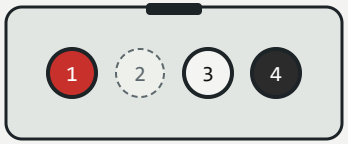
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	16 Red	w112	+12 V	Splice to: DRB102 bulkhead, engine side · pin 1; Coil 2 · pin 1; Coil 3 · pin 1; Coil 4 · pin 1	TBD
2	*spare*				
3	20 White	w120_e	Trigger (IGT)	DRB102 bulkhead, engine side · pin 9	TBD
4	16 Black	w_cop1g	Gnd	Splice to: Coil 2 · pin 4; Coil 3 · pin 4; Coil 4 · pin 4; DRB102 bulkhead, engine side · pin 4	TBD

Pin numbers: TBD · harness AB-engine

Coil 2 COIL-2 · 90980-11885

FACE SOURCED

COP 2 (1ZZ)



latch top

mating face view

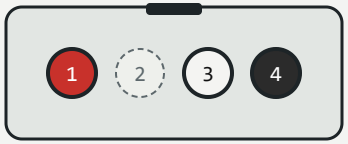
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	16 Red	w113	+12 V	Splice to: DRB102 bulkhead, engine side · pin 1; Coil 1 · pin 1; Coil 3 · pin 1; Coil 4 · pin 1	TBD
2	*spare*				
3	20 White	w121_e	Trigger (IGT)	DRB102 bulkhead, engine side · pin 10	TBD
4	16 Black	w_cop2g	Gnd	Splice to: Coil 1 · pin 4; Coil 3 · pin 4; Coil 4 · pin 4; DRB102 bulkhead, engine side · pin 4	TBD

Pin numbers: TBD · harness AB-engine

Coil 3 COIL-3 · 90980-11885

FACE SOURCED

COP 3 (1ZZ)



latch top

mating face view

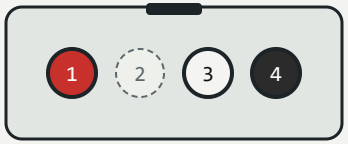
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	16 Red	w114	+12 V	Splice to: DRB102 bulkhead, engine side · pin 1; Coil 1 · pin 1; Coil 2 · pin 1; Coil 4 · pin 1	TBD
2	*spare*				
3	20 White	w122_e	Trigger (IGT)	DRB102 bulkhead, engine side · pin 11	TBD
4	16 Black	w_cop3g	Gnd	Splice to: Coil 1 · pin 4; Coil 2 · pin 4; Coil 4 · pin 4; DRB102 bulkhead, engine side · pin 4	TBD

Pin numbers: TBD · harness AB-engine

Coil 4 COIL-4 · 90980-11885

FACE SOURCED

COP 4 (1ZZ)



latch top

mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	16 Red	w115	+12 V	Splice to: DRB102 bulkhead, engine side · pin 1; Coil 1 · pin 1; Coil 2 · pin 1; Coil 3 · pin 1	TBD
2	*spare*				
3	20 White	w123_e	Trigger (IGT)	DRB102 bulkhead, engine side · pin 12	TBD
4	16 Black	w_cop4g	Gnd	Splice to: Coil 1 · pin 4; Coil 2 · pin 4; Coil 3 · pin 4; DRB102 bulkhead, engine side · pin 4	TBD

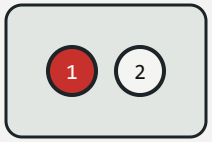
Pin numbers: TBD · harness AB-engine

Boost Control Solenoid

BOOST-SOL · 12052641

FACE NOT SOURCED

Boost Control Solenoid



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	w219	+12 V	Splice to: DRB102 bulkhead, engine side · pin 30; Flex Fuel Sensor · pin A	12084200
2	20 White	w103_e	Drive (-)	DRB102 bulkhead, engine side · pin 58	12084200

Pin numbers: TBD · harness AB-engine

ECT Sensor

ECT · 90980-11062

FACE SOURCED

ECT Sensor



latch top

mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	w201_e	Signal	DRB102 bulkhead, engine side · pin 70	8100-0461
2	20 Blue	w4	Gnd	Splice to: DRB102 bulkhead, engine side · pin 34; Bosch 74.5mm ETB · pin 2; IAT Sensor (Manifold) · pin 2	8100-0461

Pin numbers: TBD · harness AB-engine

IAT Sensor (Manifold) IAT · 90980-11062

FACE SOURCED

IAT Sensor (Manifold)



latch top

mating face view

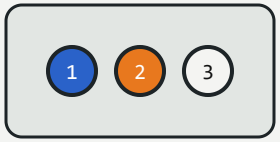
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	w202_e	Signal	DRB102 bulkhead, engine side · pin 33	8100-0461
2	20 Blue	w6	Gnd	Splice to: DRB102 bulkhead, engine side · pin 34; ECT Sensor · pin 2; Bosch 74.5mm ETB · pin 2	8100-0461

Pin numbers: TBD · harness AB-engine

Oil Pressure Sensor `OIL-P · 354404`

FACE NOT SOURCED

Oil Pressure Sensor



layout not sourced: numbered list only, match the moulded numbers

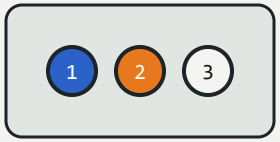
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	`w30`	Gnd	Splice to: DRB102 bulkhead, engine side · pin 29; Coolant Pressure Sensor · pin A; Charge-Pipe IAT Sensor · pin 2; Oil Temp Sensor · pin 2	`in kit 354404`
2	20 Orange	`w_5v_brail_e_c45`	+5 V	DRB102 bulkhead, engine side · pin 27	`in kit 354404`
3	20 White	`w29_e`	Signal	DRB102 bulkhead, engine side · pin 28	`in kit 354404`

Pin numbers: device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Oil pressure pin table) · harness AB-engine

Fuel Pressure Sensor FUEL-P · 354404

FACE NOT SOURCED

Fuel Pressure Sensor



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	w33	Gnd	Splice to: DRB102 bulkhead, engine side · pin 15; Gearbox speed sensor (VSS, Toyota 83181-20040) · pin 2	in kit 354404
2	20 Orange	w_5v_brail_e_c46	+5 V	DRB102 bulkhead, engine side · pin 13	in kit 354404
3	20 White	w32_e	Signal	DRB102 bulkhead, engine side · pin 14	in kit 354404

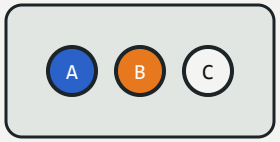
Pin numbers: device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Fuel pressure pin table) · harness AB-engine

Coolant Pressure Sensor

`CLNT-P · pigtail supplied with the sensor (owned)`

FACE NOT SOURCED

Coolant Pressure Sensor



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
A	20 Blue	`w36`	Gnd	Splice to: DRB102 bulkhead, engine side · pin 29; Oil Pressure Sensor · pin 1; Charge-Pipe IAT Sensor · pin 2; Oil Temp Sensor · pin 2	`solder splice onto the pigtail black lead`
B	20 Orange	`w_5v_brail_e_c47`	+5 V	DRB102 bulkhead, engine side · pin 25	`solder splice onto the pigtail red lead`
C	20 White	`w35_e`	Signal	DRB102 bulkhead, engine side · pin 26	`solder splice onto the pigtail green lead`

Pin numbers: Dan's sensor sheet + seller connection guide (Dan 2026-10-30): A / plug 1 GND black, B / plug 2 supply red, C / plug 3 output green · harness AB-engine

Starter solenoid S (50) terminal, 1-way plug

STARTER-TRIGGER · 90980-11400

FACE SOURCED

Starter solenoid S (50) terminal, 1-way plug



mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	12 Red	w_strl_e	Solenoid S (start)	DRB102 bulkhead, engine side · pin 3	in kit CONN-76058 (Sumitomo TS187 4.8 mm sealed female, 12-10 AWG)

Pin numbers: single cavity (Toyota RM1022E housing list: 90980-11400 is 1-way, 4.8 mm tab, sealed) · harness AB-engine

Reverse switch (EWD back-up SW B1)

REV-SW · 90980-11250

FACE SOURCED

Reverse switch (EWD back-up SW B1)



latch top

mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	16 Yellow	w_sw7_e	Switched +12 V (NO)	DRB102 bulkhead, engine side · pin 78	8100-0461
2	16 Red	w_rev_feed_e	+12 V (COM)	DRB102 bulkhead, engine side · pin 77	8100-0461

Pin numbers: TBD · harness AB-engine

Oil Temp Sensor

OIL-T · 90980-11062

FACE SOURCED

Oil Temp Sensor



latch top

mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	w203_e	Signal	DRB102 bulkhead, engine side · pin 61	8100-0461
2	20 Blue	w8	Gnd	Splice to: Charge-Pipe IAT Sensor · pin 2; DRB102 bulkhead, engine side · pin 29; Coolant Pressure Sensor · pin A; Oil Pressure Sensor · pin 1	8100-0461

Pin numbers: TBD · harness AB-engine

Charge-Pipe IAT Sensor

CHARGE-IAT · 90980-11062

FACE SOURCED

Charge-Pipe IAT Sensor



latch top

mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	w204_e	Signal	DRB102 bulkhead, engine side · pin 59	8100-0461
2	20 Blue	w10	Gnd	Splice to: Oil Temp Sensor · pin 2; DRB102 bulkhead, engine side · pin 29; Coolant Pressure Sensor · pin A; Oil Pressure Sensor · pin 1	8100-0461

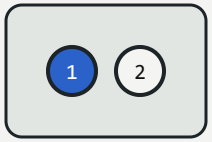
Pin numbers: TBD · harness AB-engine

Knock Sensor 1

KNOCK-1 · 1 928 403 874

FACE NOT SOURCED

Knock Sensor 1



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	cab_knock1_c1	Signal	DRB102 bulkhead, engine side · pin 71	BDK 2.8
2	20 White	cab_knock1_c2	Gnd	DRB102 bulkhead, engine side · pin 72	BDK 2.8

Pin numbers: TBD · harness AB-engine

Gearbox speed sensor (VSS, Toyota 83181-20040)

VSS · 90980-11143

FACE SOURCED

Gearbox speed sensor (VSS, Toyota 83181-20040)



mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Orange	w_vss_8v	+8 V	DRB102 bulkhead, engine side · pin 67	8100-0461
2	20 Blue	w_vss_gnd	Gnd	Splice to: DRB102 bulkhead, engine side · pin 15; Fuel Pressure Sensor · pin 1	8100-0461
3	20 White	w_vss_sig_e	Signal	DRB102 bulkhead, engine side · pin 68	8100-0461

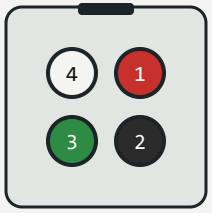
Pin numbers: Toyota 3-wire speed sensor pinout (wilbo666 wiki; Daniel 2026-09-28) · harness AB-engine

Link CAN-Lambda

CAN-LAMBDA · DTM06-4S + WM-4S

FACE SOURCED

Link CAN-Lambda



latch top

wire side (where the contacts go in) view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	16 Red	w_lam12_be	+12 V	Splice to: DRB102 bulkhead, engine side · pin 95	0462-005-2031
2	16 Black	w_lamg_be	Gnd	Splice to: DRB102 bulkhead, engine side · pin 96	0462-005-2031
3	20 Green	wc_l_lam_pin	CAN L	Splice to: DRB102 bulkhead, engine side · pin 80; 120 ohm CAN terminator plug, Lambda end · pin 2	0462-201-2031
4	20 White	wc_h_lam_pin	CAN H	Splice to: DRB102 bulkhead, engine side · pin 79; 120 ohm CAN terminator plug, Lambda end · pin 1	0462-201-2031

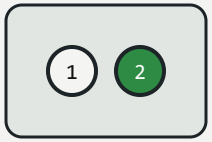
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness AB-engine

120 ohm CAN terminator plug, Lambda end

CAN-TERM-END · AT06-2S-RD01 + AW2S

FACE NOT SOURCED

CAN-Lambda end 120 ohm terminator plug



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	wc_term3	CAN H	Splice to: DRB102 bulkhead, engine side · pin 79; Link CAN-Lambda · pin 4	AT62-201-1631
2	20 Green	wc_term4	CAN L	Splice to: DRB102 bulkhead, engine side · pin 80; Link CAN-Lambda · pin 3	AT62-201-1631

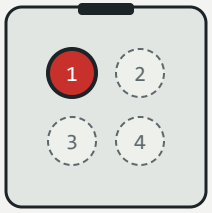
Pin numbers: TBD · harness AB-engine

MR-S EHPS pump plug C

MRS-C · 90980-10942

FACE SOURCED

MR-S EHPS pump plug C



mating face view

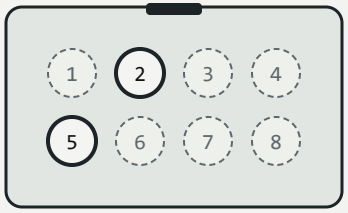
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	18 Red	w_hv_mrs_en_e	+12 V (IG)	DRB102 bulkhead, engine side · pin 83	82998-12440
2	*spare*				
3	*spare*				
4	*spare*				

Pin numbers: MR-S (ZZW30) EWD power steering page (Dan 2026-10-30): P6 1 MH, 2 PGND; P7 1 IDUP, 2 SPD, 3 TC, 5 MRLY, 6 EFI, 7 TS; P8 1 IGB, 2 SIL, 4 WL · harness AB-engine

MR-S EHPS pump plug B `MRS-B · 90980-10897`

FACE SOURCED

MR-S EHPS pump plug B



mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	*spare*				
2	☐ 20 White	`w_mrs_e`	Speed signal	DRB102 bulkhead, engine side · pin 84	`82998-12440`
3	*spare*				
4	*spare*				
5	☐ 18 White	`w_eps_rly_e`	Relay coil drive (-)	DRB102 bulkhead, engine side · pin 85	`82998-12440`
6	*spare*				
7	*spare*				
8	*spare*				

Pin numbers: MR-S (ZZW30) EWD power steering page (Dan 2026-10-30): P6 1 MH, 2 PGND; P7 1 IDUP, 2 SPD, 3 TC, 5 MRLY, 6 EFI, 7 TS; P8 1 IGB, 2 SIL, 4 WL · harness AB-engine

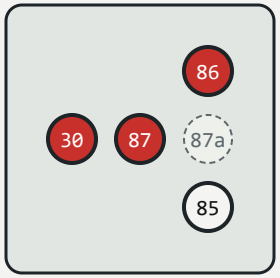
CabinPower

LOGIC fuse box relay M1 (low beam (both headlamps))

K-HL · 5-1393292-8

FACE SOURCED

B2-M1 Low-beam relay, both headlamps (seated in the LOGIC box M1)



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	14 Red	w_hl_30_cp	Low-beam relay feed	Splice to: LOGIC fuse box · fuse BUS1-2; LOGIC fuse box relay M1 (low beam (both headlamps)) · terminal 86; DRL cut relay RLY-DRL (Dan N90 a) · pin 86	42281-1
85	18 White	w_hl_85_cp	Low-beam relay coil (-)	Splice to: OEM cowl plug IE1 pin 2 (LH kick), R-Y wire; DRL cut relay RLY-DRL (Dan N90 a) · pin 85	280919-2
86	16 Red	w_hl_86_cp	Low-beam relay coil +	Splice to: LOGIC fuse box · fuse BUS1-2; LOGIC fuse box relay M1 (low beam (both headlamps)) · terminal 30; DRL cut relay RLY-DRL (Dan N90 a) · pin 86	280919-2
87	14 Red	w_hl_on_cp	Low beam +12 V out	Splice to: LOGIC fuse box relay M2 (high beam (both headlamps)) · terminal 86; LH headlamp H1 pin 1; RH headlamp H2 pin 1	42281-1
87a	*spare*				

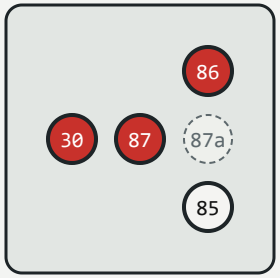
Pin numbers: ISO 7588 relay terminal numbers · harness CabinPower

LOGIC fuse box relay M3 (horn)

K-HORN · 4-1904124-3

FACE SOURCED

B2-M3 Horn relay (seated in the LOGIC box M3)



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	14 Red	w_horn_30_cp	Horn relay feed	Splice to: LOGIC fuse box · fuse BUS1-4; OEM IE1 pin 12 (left kick, cowl side); LOGIC fuse box relay M3 (horn) · terminal 86	42281-1
85	18 White	w_horn_85_cp	Horn relay coil (-)	OEM cowl plug IE1 pin 19 (LH kick), G wire	280919-2
86	16 Red	w_horn_86_cp	Horn relay coil +	Splice to: LOGIC fuse box · fuse BUS1-4; OEM IE1 pin 12 (left kick, cowl side); LOGIC fuse box relay M3 (horn) · terminal 30	280919-2
87	14 Red	w_pmu_horn_cp	Horn +12 V out	Splice to: Horn LH (EWD H3) · pin 1; Horn RH (EWD H4) · pin 1	42281-1
87a	*spare*				

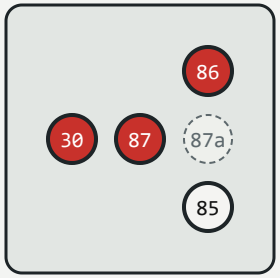
Pin numbers: ISO 7588 relay terminal numbers · harness CabinPower, EngineRoom-C

LOGIC fuse box relay M2 (high beam (both headlamps))

K-HR · 5-1393292-8

FACE SOURCED

B2-M2 High-beam relay, both headlamps (seated in the LOGIC box M2)



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

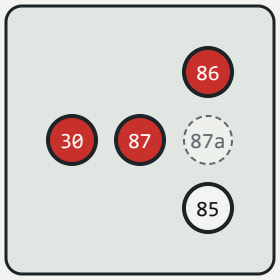
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	14 Red	w_hr_30_cp	High-beam relay feed	Splice to: LOGIC fuse box · fuse BUS1-3	42281-1
85	18 White	w_hl_low	High-beam relay coil (-)	OEM IE1 pin 18 (left kick, cowl side)	280919-2
86	16 Red	w_hr_86_cp	High-beam relay coil +	Splice to: LOGIC fuse box relay M1 (low beam (both headlamps)) · terminal 87; LH headlamp H1 pin 1; RH headlamp H2 pin 1	280919-2
87	14 Red	w_hr_on_cp	High beam +12 V out	Splice to: Dash high-beam LED feed resistor 220 ohm 1 W · pin 1; RH headlamp H2 pin 2; LH headlamp H1 pin 2	42281-1
87a	*spare*				

Pin numbers: ISO 7588 relay terminal numbers · harness CabinPower

LOGIC fuse box relay M8 (DRL (both headlamps)) K-DRL · 4-1904124-2

FACE SOURCED

B2-M8 DRL relay (seated in the LOGIC box M8, a spare seat) - coil+ (86) from LOGIC BUS2-1 DRL 7.5 A, coil- (85) through the DRL switch; feeds the DRL cut relay RLY-DRL (30)



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

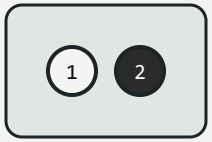
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	18 Red	w_drl_30_cp	DRL relay feed	Splice to: LOGIC fuse box · fuse BUS2-1; LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 86; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 86	42281-1
85	20 White	w_drl_85_cp	DRL relay coil (-)	Splice to: DRL switch · pin 1; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 85; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 87	280919-2
86	18 Red	w_drl_86_cp	DRL relay coil +	Splice to: LOGIC fuse box · fuse BUS2-1; LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 30; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 86	280919-2
87	16 Red	w_drl_87_cp	DRL +12 V out	DRL cut relay RLY-DRL (Dan N90 a) · pin 30	42281-1

Pin numbers: ISO 7588 relay terminal numbers · harness CabinPower

DRL switch DRL-SW · TBD

FACE NOT SOURCED

DRL switch - SPST on / off, signal only (DRL relay coil and pop-up diode D2 to ground); left of the steering wheel (Dan 2026-11-08), type TBD (Dan)



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	☐ 20 White	w_drl_sw_cp	DRL relay coil and pop-up UP	Splice to: LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 85; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 85; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 87	TBD
2	■ 20 Black	w_drl_sw_gnd_cp	Ground	OEM retract control relay R8 pin 12 (behind the LH dash), W-B ground lead	TBD

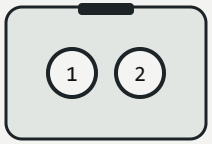
Pin numbers: switch terminals 1 / 2 (as drawn) · harness CabinPower

Pop-up diode D1 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug)

`D-R8-1 · DT06-2S + W2S;DT04-2P-RT01`

FACE SOURCED

Pop-up diode D1 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug) - R8 pin 8 to the stalk side (cowl splice I2), behind the LH dash



`latch top`

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	☐ 20 White	`w_r8_d1_a_cp`	D1 anode (R8 pin 8 side)	Splice to: OEM retract control relay R8 pin 8 (behind the LH dash), R-Y lead cut near R8: R8 side; Pop-up diode D2 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug) · pin 1	`0462-201-1631`
2	☐ 20 White	`w_r8_d1_k_cp`	D1 cathode (stalk side)	OEM R-Y lead from cowl splice I2, cut near retract control relay R8 pin 8: I2 side (behind the LH dash)	`0462-201-1631`

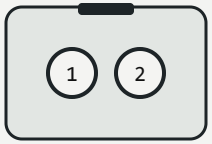
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing); which cavity the MUR460 anode sits on: TBD by the TE drawing of DT04-2P-RT01 (swap the two wires if it is the other one) · harness CabinPower

Pop-up diode D2 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug)

`D-R8-2 · DT06-2S + W2S;DT04-2P-RT01`

FACE SOURCED

Pop-up diode D2 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug) - R8 pin 8 to the DRL switch, behind the LH dash



`latch top`

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	☐ 20 White	`w_r8_d2_a_cp`	D2 anode (R8 pin 8 side)	Splice to: OEM retract control relay R8 pin 8 (behind the LH dash), R-Y lead cut near R8: R8 side; Pop-up diode D1 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug) · pin 1	`0462-201-1631`
2	☐ 18 White	`w_r8_d2_k_cp`	D2 cathode (DRL switch side)	LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 30	`0462-201-1631`

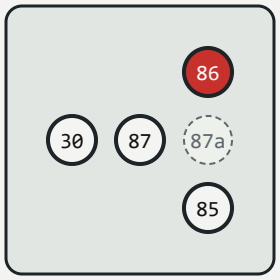
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing); which cavity the MUR460 anode sits on: TBD by the TE drawing of DT04-2P-RT01 (swap the two wires if it is the other one) · harness CabinPower

LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8))

K-POD · 4-1904124-3

FACE SOURCED

B2-M10 Pop-up key-on relay (seated in the LOGIC box M10, a spare seat) - coil+ (86) from LOGIC BUS2-1 DRL 7.5 A (key on), coil- (85) through the DRL switch; 30 / 87 put pop-up diode D2 on the DRL switch only with the key on (Dan N93 a)



mating face view. Relay terminal numbers mapped to ISO 85 / 86 / 30 / 87 / 87a (1 = 86, 2 = 85, 3 = 30, 4 = 87a, 5 = 87).

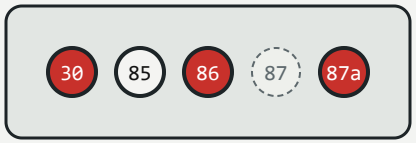
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	☐ 18 White	w_r8_d2_k_cp	Pop-up UP command (D2), switched by K-POD	Pop-up diode D2 (DEUTSCH DT04-2P-RT01 diode receptacle, MUR460, mated to a DT06-2S plug) · pin 2	42281-1
85	☐ 20 White	w_pod_85_cp	Pop-up relay coil (-)	Splice to: LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 85; DRL switch · pin 1; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 87	280919-2
86	■ 18 Red	w_pod_86_cp	Pop-up relay coil +	Splice to: LOGIC fuse box · fuse BUS2-1; LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 30; LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 86	280919-2
87	☐ 18 White	w_pod_87_cp	Pop-up UP command to the DRL switch (key on)	Splice to: LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 85; DRL switch · pin 1; LOGIC fuse box relay M10 (pop-up key-on (DRL switch to R8 pin 8)) · terminal 85	42281-1
87a	*spare*				

Pin numbers: ISO 7588 relay terminal numbers · harness CabinPower

DRL cut relay RLY-DRL (Dan N90 a) `RLY-DRL · 12110539 + 12110540;301-1C-C-R1-U01-12VDC`

FACE NOT SOURCED

DRL cut relay RLY-DRL (Dan N90 a) - Song Chuan 301-1C-C-R1-U01-12VDC (ISO 280 micro, SPDT, resistor) in a Delphi Metri-Pack 280 5-way socket 12110539, beside the LOGIC box (outside it): coil on R-Y, 87a (NC) passes the DRL feed only while the head lamps are off



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
30	16 Red	w_drl_87_cp	DRL feed in	LOGIC fuse box relay M8 (DRL (both headlamps)) · terminal 87	12129409-L
85	16 White	w_drlc_85_cp	DRL cut relay coil (-)	Splice to: LOGIC fuse box relay M1 (low beam (both headlamps)) · terminal 85; OEM cowl plug IE1 pin 2 (LH kick), R-Y wire	12129409-L
86	16 Red	w_drlc_86_cp	DRL cut relay coil +	Splice to: LOGIC fuse box · fuse BUS1-2; LOGIC fuse box relay M1 (low beam (both headlamps)) · terminal 30; LOGIC fuse box relay M1 (low beam (both headlamps)) · terminal 86	12129409-L
87	*spare*				
87a	16 Red	w_pmu_drl	DRL +12 V out (head lamps off)	Splice to: LH headlamp H1 white DRL lead; RH headlamp H2 white DRL lead	12129409-L

Pin numbers: ISO 7588 relay terminal numbers (Song Chuan 301 series data sheet) · harness CabinPower, EngineRoom-C

EngineRoom-C

MR-S EHPS pump plug A

MRS-A · 90980-12068

FACE SOURCED

MR-S EHPS pump plug A



latch top

mating face view

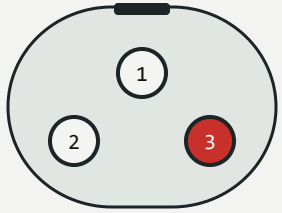
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	8 Red	w_eps_pwr_bc	EHPS pump +12 V	EPS pump relay (beside the POWER fuse box) · terminal 87	82998-12500
2	8 Black	w_eps_gnd	Gnd	Ground stud EA, RH front fender (EWD ground point EA)	82998-12500

Pin numbers: MR-S (ZZW30) EWD power steering page (Dan 2026-10-30): P6 1 MH, 2 PGND; P7 1 IDUP, 2 SPD, 3 TC, 5 MRLY, 6 EFI, 7 TS; P8 1 IGB, 2 SIL, 4 WL · harness EngineRoom-C

Alternator G2 ALT-PLUG · 90980-10341

FACE SOURCED

Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout)



latch top

mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	w_alt_s_c	S (battery sense)	LOGIC fuse box · fuse BUS1-9	TBD
2	20 White	w_alt_l_c	L (charge lamp)	OEM IG2 pin 16 (under the ECU, cowl side)	TBD
3	16 Red	w_ig_ks_out	Alt IG excite	Kill switch · pin ALT2	TBD

Pin numbers: TBD · harness EngineRoom-C

Uprated condenser fan (GND)

FAN-COND-GND · DTHD06-1-8S

FACE SOURCED

Uprated condenser fan (GND)



mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	10 Black	w_fan2_gnd	Gnd	Ground stud EA, RH front fender (EWD ground point EA)	0462-203-08141

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness EngineRoom-C

Uprated radiator fan (+) FAN-RAD · DTHD06-1-8S

FACE SOURCED

Uprated radiator fan (+)



mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	10 Red	w_fan_pwr	+12 V	POWER fuse box relay R4 (radiator fan) · terminal 30	0462-203-08141

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness EngineRoom-C

Uprated condenser fan (+) FAN-COND · DTHD06-1-8S

FACE SOURCED

Uprated condenser fan (+)



mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	10 Red	w_fan2_pwr	+12 V	POWER fuse box relay R5 (condenser fan) · terminal 30	0462-203-08141

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness EngineRoom-C

Horn LH (EWD H3) HORN-LH · DT06-2S-CE05 + W2S

FACE SOURCED

Horn LH (EWD H3) - custom DT 2-way



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	14 Red	w_horn_lh	+12 V	Splice to: LOGIC fuse box relay M3 (horn) · terminal 87; Horn RH (EWD H4) · pin 1	0462-209-1631
2	14 Black	w_horn_lh_gs	Gnd	Splice to: Ground stud EA, RH front fender (EWD ground point EA); Horn RH (EWD H4) · pin 2	0462-209-1631

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness EngineRoom-C

Horn RH (EWD H4) HORN-RH · DT06-2S-CE05 + W2S

FACE SOURCED

Horn RH (EWD H4) - custom DT 2-way



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	14 Red	w_horn_rh	+12 V	Splice to: LOGIC fuse box relay M3 (horn) · terminal 87; Horn LH (EWD H3) · pin 1	0462-209-1631
2	14 Black	w_horn_rh_gs	Gnd	Splice to: Ground stud EA, RH front fender (EWD ground point EA); Horn LH (EWD H3) · pin 2	0462-209-1631

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness EngineRoom-C

Uprated radiator fan (GND) FAN-RAD-GND · DTHD06-1-8S

FACE SOURCED

Uprated radiator fan (GND)



mating face view

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	10 Black	w_fan_gnd	Gnd	Ground stud EA, RH front fender (EWD ground point EA)	0462-203-08141

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness EngineRoom-C

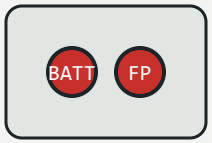
HeavyDC

Trunk relay box

BATT-RELAY-BLOCK · ST185-FB-BATT rev 7

FACE NOT SOURCED

Battery relay block - fuel pump relay + its fuse, trunk, passenger side between the rear wall and the battery (cad/fusebox rev 7 = models folder st185_fusebox_1101 revB, drawing ST185-FB-100B; DECISIONS R75, Z4)



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
BATT	10 Red	w_fp_feed	Battery relay box +	Kill switch · pin MAIN2	TBD
FP	10 Red	w_fp_30	Fuel pump relay feed	Trunk relay box relay K1 (fuel pump) · terminal 30	36161

Pin numbers: ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204) · harness HeavyDC

RADLOK + firewall

RADLOK-POS-CAB · RL00801-50RE

FACE NOT SOURCED

RADLOK + firewall



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	1/0 Red	w_h3_radlok_pos	Starter feed (H3)	POWER box + bar, RADLOK+ stud	INTEGRAL
1	1/0 Red	w_rl_pos	Starter feed (H3)	RADLOK + engine · pin 1	INTEGRAL

Pin numbers: TBD · harness HeavyDC

RADLOK RADLOK-NEG-CAB · RL00801-50BK

FACE NOT SOURCED

RADLOK - firewall



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	1/0 Black	w_h6a_radlok_neg	Engine ground (H6a)	POWER box NEG bar, RADLOK- stud	INTEGRAL
1	1/0 Black	w_rln	Engine ground (H6a)	RADLOK · pin 1	INTEGRAL

Pin numbers: TBD · harness HeavyDC

RADLOK RADLOK-NEG-ENG · RL00801-50BK

FACE NOT SOURCED

RADLOK - engine



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	1/0 Black	w_rln_blk	Engine ground (H6a)	Engine block ground stud, lug 1 (Route 1 point 5)	INTEGRAL
1	1/0 Black	w_rln	Engine ground (H6a)	RADLOK · pin 1	INTEGRAL

Pin numbers: TBD · harness HeavyDC

RADLOK + engine

RADLOK-POS-ENG · RL00801-50RE

FACE NOT SOURCED

RADLOK + engine



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	1/0 Red	w_rlp_str	Battery + to the jump bus bar	Splice to: Starter B+ stud, its one lug (Route 1 point 2); Engine-bay jump post · pin 1; Alternator B+ stud, rear RH of engine (Route 1 point 3)	INTEGRAL
1	1/0 Red	w_rl_pos	Starter feed (H3)	RADLOK + firewall · pin 1	INTEGRAL

Pin numbers: TBD · harness HeavyDC

Engine-bay jump post JUMP-POST · TBD

FACE NOT SOURCED

Engine-bay jump post



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	2 Red	w_jump	Jump post feed	Splice to: Starter B+ stud, its one lug (Route 1 point 2); RADLOK + engine · pin 1; Alternator B+ stud, rear RH of engine (Route 1 point 3)	TBD

Pin numbers: TBD · harness HeavyDC

RearFuel

Fuel Pump FUEL-PUMP · TBD

FACE NOT SOURCED

Fuel Pump



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	10 Red	w_fpump_r	Fuel pump +12 V	Trunk relay box relay K1 (fuel pump) · terminal 87	TBD
2	10 Black	w_fpg	Fuel pump ground	Ground point BL, back panel centre (EWD ground point BL)	TBD

Pin numbers: TBD · harness RearFuel

Fuel sender inline plug, sender side, trunk (IX-FUEL)

`IX-FUEL-S · DT06-2S + W2S`

FACE SOURCED

IX-FUEL inline to fuel sender (trunk LH)



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	`w_fl_sig_r`	Fuel level signal	Fuel Level Sender · pin 1	`0462-201-16141`
2	20 Blue	`w45_r`	Fuel level ground	Fuel Level Sender · pin 2	`0462-201-16141`

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness RearFuel

Fuel Level Sender FUEL-LEVEL · DT04-2P + W2P

FACE SOURCED

Fuel Level Sender - DT 2-way (Dan rev 2, 2026-10-10 - no OEM connector)



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 White	w_fl_sig_r	Fuel level signal	Fuel sender inline plug, sender side, trunk (IX-FUEL) · pin 1	0460-202-1631
2	20 Blue	w45_r	Fuel level ground	Fuel sender inline plug, sender side, trunk (IX-FUEL) · pin 2	0460-202-1631

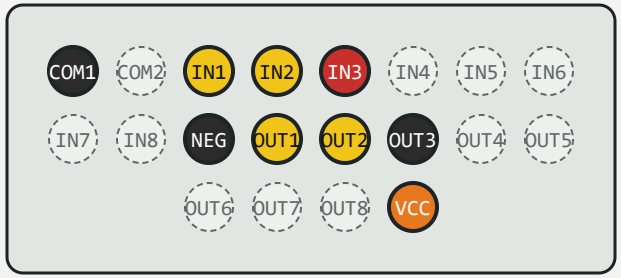
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness RearFuel

Dash

Optocoupler board, 8 channel (Dan's, generic EL814) OPTO-8 · TBD

FACE NOT SOURCED

Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
COM1	20 Black	w_opto_inc	Opto COM ground	Splice to: Cluster buck converter IN- pin (old meter spot); Pi 12-to-5 V supply IN- pin (center dash); DRB102 bulkhead, cabin side · pin 96; DRB102 bulkhead, cabin side · pin 62; DRB102 bulkhead, cabin side · pin 4; POWER box NEG bar, GND stud, ring 3 of 3; OEM A/C amplifier A15; ECUMaster CSB3 · pin 2; OEM combination switch plug C16 (B) pin 17 (cruise control switch), BR wire	screw terminal (on the board)
COM2	*spare*				
IN1	16 Yellow	w_sw6	Brake on +12 V	OEM stop lamp switch S9 pin 1 (pedal), G-W wire	screw terminal (on the board)
IN2	16 Yellow	w_sw7_c	Reverse to opto in 2	Splice to: DRB102 bulkhead, cabin side · pin 78; OEM IG2 pin 13 (under the ECU, cowl side)	screw terminal (on the board)
IN3	20 Red	w_hl_sense_in	Lights on +12 V	OEM dimmer rheostat R10 pin 1 (left of the steering wheel), green TAIL	screw terminal (on the board)
IN4	*spare*				
IN5	*spare*				
IN6	*spare*				
IN7	*spare*				
IN8	*spare*				
NEG	20 Black	w_opto_outc	Opto out ground	ECUMaster CSB3 · pin 4	screw terminal (on the board)
OUT1	20 Yellow	w_opto_out1	Brake to CSB3 S7	ECUMaster CSB3 · pin 21	screw terminal (on the board)

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
OUT2	20 Yellow	w_opto_out2	Reverse to CSB3 S8	ECUMaster CSB3 · pin 22	screw terminal (on the board)
OUT3	20 Black	w_hl_sense_out	Headlight sense to cluster	Centre cluster headlight-sense input (J8 pin 15, GPIO28, active-low)	screw terminal (on the board)
OUT4	*spare*				
OUT5	*spare*				
OUT6	*spare*				
OUT7	*spare*				
OUT8	*spare*				
VCC	20 Orange	w_opto_vcc	Opto +5 V	ECUMaster CSB3 · pin 3	screw terminal (on the board)

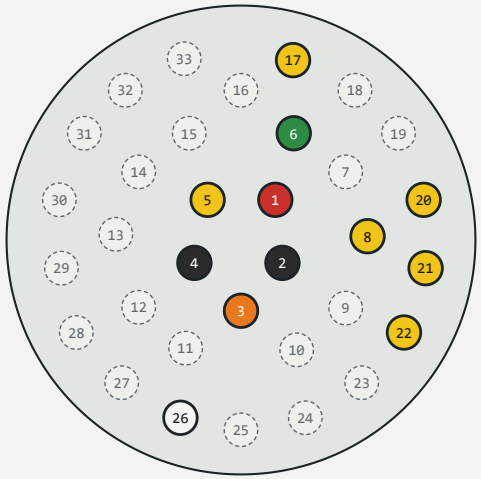
Pin numbers: terminal names printed on the board · harness Dash

ECUMaster CSB3

CSB3 · HD36-24-33SE

FACE SOURCED

ECUMaster CSB3 - Power/CAN/Switch I/O



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	wc_12v_csb	CSB3 +12 V	Splice to: LOGIC fuse box · fuse BUS2-5	0462-201-2031
2	20 Black	w_csbg	CSB3 ground	Splice to: Cluster buck converter IN- pin (old meter spot); Pi 12-to-5 V supply IN- pin (center dash); DRB102 bulkhead, cabin side · pin 96; DRB102 bulkhead, cabin side · pin 62; DRB102 bulkhead, cabin side · pin 4; POWER box NEG bar, GND stud, ring 3 of 3; OEM A/C amplifier A15; OEM combination switch plug C16 (B) pin 17 (cruise control switch), BR wire; Optocoupler board, 8 channel (Dan's, generic EL814) · pin COM1	0462-201-2031
3	20 Orange	w_opto_vcc	Opto +5 V	Optocoupler board, 8 channel (Dan's, generic EL814) · pin VCC	0462-201-2031
4	20 Black	w_opto_outc	Opto out ground	Optocoupler board, 8 channel (Dan's, generic EL814) · pin NEG	0462-201-2031
5	20 Yellow	wc_h_csb	CAN H to CSB3	Splice to: DRB102 bulkhead, cabin side · pin 79; Link G4X plug C, CAN / comms port (generic round 6-pin plug, centre dash) · pin 3; Cluster CAN board CANH + CANL pins (old meter spot); Pi USB-CAN adapter CANH + CANL pins (center dash)	0462-201-2031
6	20 Green	wc_l_csb	CAN L to CSB3	Splice to: DRB102 bulkhead, cabin side · pin 80; Link G4X plug C, CAN / comms port (generic round 6-pin plug, centre dash) · pin 4; Cluster CAN board CANH + CANL pins (old meter spot); Pi USB-CAN adapter CANH + CANL pins (center dash)	0462-201-2031
7	*spare*				

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
8	20 Yellow	w_cru_ladder_out	Cruise ladder	Splice to: OEM combination switch plug C16 (B) pin 5 (cruise control switch), W wire; Cruise resistor on the CSB3 · pin 2	0462-201-2031
9	*spare*				
10	*spare*				
11	*spare*				
12	*spare*				
13	*spare*				
14	*spare*				
15	*spare*				
16	*spare*				
17	20 Yellow	w_sw2	Cruise MAIN on	OEM combination switch plug C16 (B) pin 15 (cruise control switch), Y-B wire	0462-201-2031
18	*spare*				
19	*spare*				
20	20 Yellow	w_sw5	Clutch pedal sw	OEM cruise control clutch switch C17 pin 2, B wire	0462-201-2031
21	20 Yellow	w_opto_out1	Brake to CSB3 S7	Optocoupler board, 8 channel (Dan's, generic EL814) · pin OUT1	0462-201-2031
22	20 Yellow	w_opto_out2	Reverse to CSB3 S8	Optocoupler board, 8 channel (Dan's, generic EL814) · pin OUT2	0462-201-2031
23	*spare*				
24	*spare*				
25	*spare*				
26	20 White	w_oil_k	Oil LED sink (CSB3)	Oil pressure LED · pin 2	0462-201-2031
27	*spare*				
28	*spare*				
29	*spare*				
30	*spare*				
31	*spare*				

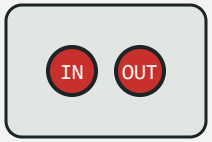
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
32	*spare*				
33	*spare*				

Pin numbers: TBD · harness Dash

RealDash Pi 5 A inline fuse `F-PI · 0FHM0003ZXJB + ATM-5`

FACE NOT SOURCED

RealDash Pi inline fuse 5 A at the CAN joins by ECU plug C (Littelfuse MINI FHM in-line holder 0FHM0003ZXJB, 16 AWG leads, with a 5 A MINI fuse)



layout not sourced: numbered list only, match the moulded numbers

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
IN	16 Red	w_pi12	J/B No.1 CIG & RADIO 15 A (ACC)	OEM cowl plug II1 pin 3 (instrument panel centre, cowl side), GR	solder splice onto the holder lead
OUT	20 Red	wc_12v_rd	RealDash Pi +12 V (after the 5 A inline fuse)	Pi 12-to-5 V supply IN+ pin (center dash)	solder splice onto the holder lead

Pin numbers: Littelfuse INLINE MINI Splashproof Fuse Holder datasheet: 0FHM0003ZXJB, 16 AWG GXL leads 93 mm each side, 20 A, 32 V DC, takes MINI blade fuses · harness Dash

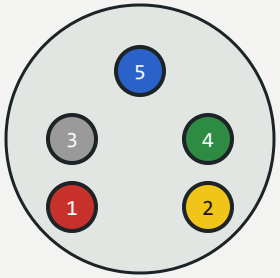
WheelSpeed-Rear

VRC rear OUT plug

VRC-R-OUT · LP-16-J05PE-02-001

FACE SOURCED

VRC rear OUT plug



mating face view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	cab_rout_5v	VRC-R +5V	Rear wheel-speed inline plug, sensor side (IX-WS-R) · pin 1	solder cup (LP-16, S16)
2	20 Yellow	cab_rout_gnd	VRC-R GND	Rear wheel-speed inline plug, sensor side (IX-WS-R) · pin 2	solder cup (LP-16, S16)
3	Shield	cab_rout_sh	WS-R screen drain	Rear wheel-speed inline plug, sensor side (IX-WS-R) · pin 5	solder cup (LP-16, S16)
4	20 Green	cab_rout_l	Rear left wheel spd	Rear wheel-speed inline plug, sensor side (IX-WS-R) · pin 3	solder cup (LP-16, S16)
5	20 Blue	cab_rout_r	Rear right wheel spd	Rear wheel-speed inline plug, sensor side (IX-WS-R) · pin 4	solder cup (LP-16, S16)

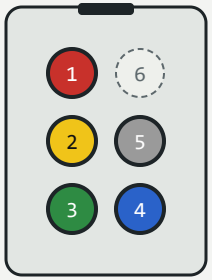
Pin numbers: CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19) · harness WheelSpeed-Rear

Rear wheel-speed inline plug, sensor side (IX-WS-R)

`IX-WS-R-P · DT04-6P + W6P`

FACE SOURCED

IX-WS-R



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	`cab_rout_5v`	VRC-R +5V	VRC rear OUT plug · pin 1	`0460-202-1631`
2	20 Yellow	`cab_rout_gnd`	VRC-R GND	VRC rear OUT plug · pin 2	`0460-202-1631`
3	20 Green	`cab_rout_l`	Rear left wheel spd	VRC rear OUT plug · pin 4	`0460-202-1631`
4	20 Blue	`cab_rout_r`	Rear right wheel spd	VRC rear OUT plug · pin 5	`0460-202-1631`
5	Shield	`cab_rout_sh`	WS-R screen drain	VRC rear OUT plug · pin 3	`0460-202-1631`
6	*spare*				

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness WheelSpeed-Rear

ABS Wheel Speed RL

WSS-RL · DT06-2S-CE05 + W2S

FACE SOURCED

ABS Wheel Speed RL



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	cab_rl_p	RL wheel speed +	VRC rear IN plug · pin 1	0462-201-16141
2	20 White	cab_rl_n	RL wheel speed -	VRC rear IN plug · pin 2	0462-201-16141

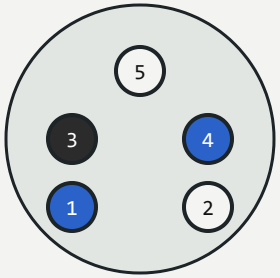
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness WheelSpeed-Rear

VRC rear IN plug

`VRC-R-IN · LP-16-J05PE-02-001`

FACE SOURCED

VRC rear IN plug



mating face view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	`cab_rl_p`	RL wheel speed +	ABS Wheel Speed RL · pin 1	`solder cup (LP-16, S16)`
2	20 White	`cab_rl_n`	RL wheel speed -	ABS Wheel Speed RL · pin 2	`solder cup (LP-16, S16)`
3	20 Black	`w_ws_r_shield`	Rear sensor shields	Splice to: Shield, not connected at the device end	`solder cup (LP-16, S16)`
4	20 Blue	`cab_rr_p`	RR wheel speed +	ABS Wheel Speed RR · pin 1	`solder cup (LP-16, S16)`
5	20 White	`cab_rr_n`	RR wheel speed -	ABS Wheel Speed RR · pin 2	`solder cup (LP-16, S16)`

Pin numbers: CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19) · harness WheelSpeed-Rear

ABS Wheel Speed RR `WSS-RR · DT06-2S-CE05 + W2S`

FACE SOURCED

ABS Wheel Speed RR



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	`cab_rr_p`	RR wheel speed +	VRC rear IN plug · pin 4	`0462-201-16141`
2	20 White	`cab_rr_n`	RR wheel speed -	VRC rear IN plug · pin 5	`0462-201-16141`

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness WheelSpeed-Rear

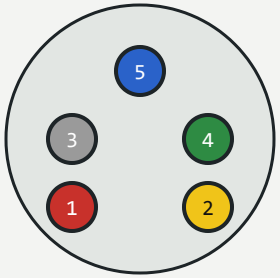
WheelSpeed-Front

VRC front OUT plug

VRC-F-OUT · LP-16-J05PE-02-001

FACE SOURCED

VRC front OUT plug



mating face view · source

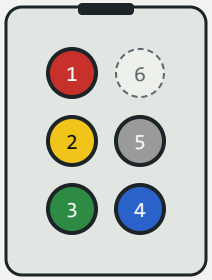
PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	cab_fout_5v	VRC-F +5V	Front wheel-speed inline plug, sensor side (IX-WS-F) · pin 1	solder cup (LP-16, S16)
2	20 Yellow	cab_fout_gnd	VRC-F GND	Front wheel-speed inline plug, sensor side (IX-WS-F) · pin 2	solder cup (LP-16, S16)
3	Shield	cab_fout_sh	WS-F screen drain	Front wheel-speed inline plug, sensor side (IX-WS-F) · pin 5	solder cup (LP-16, S16)
4	20 Green	cab_fout_l	FL conditioned out	Front wheel-speed inline plug, sensor side (IX-WS-F) · pin 3	solder cup (LP-16, S16)
5	20 Blue	cab_fout_r	FR conditioned out	Front wheel-speed inline plug, sensor side (IX-WS-F) · pin 4	solder cup (LP-16, S16)

Pin numbers: CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19) · harness WheelSpeed-Front

Front wheel-speed inline plug, sensor side (IX-WS-F) IX-WS-F-P · DT04-6P + W6P

FACE SOURCED

IX-WS-F



wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Red	cab_fout_5v	VRC-F +5V	VRC front OUT plug · pin 1	0460-202-1631
2	20 Yellow	cab_fout_gnd	VRC-F GND	VRC front OUT plug · pin 2	0460-202-1631
3	20 Green	cab_fout_l	FL conditioned out	VRC front OUT plug · pin 4	0460-202-1631
4	20 Blue	cab_fout_r	FR conditioned out	VRC front OUT plug · pin 5	0460-202-1631
5	Shield	cab_fout_sh	WS-F screen drain	VRC front OUT plug · pin 3	0460-202-1631
6	*spare*				

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness WheelSpeed-Front

ABS Wheel Speed FL

`WSS-FL · DT06-2S-CE05 + W2S`

FACE SOURCED

ABS Wheel Speed FL



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	`cab_fl_p`	FL wheel speed +	VRC front IN plug · pin 1	`0462-201-16141`
2	20 White	`cab_fl_n`	FL wheel speed -	VRC front IN plug · pin 2	`0462-201-16141`

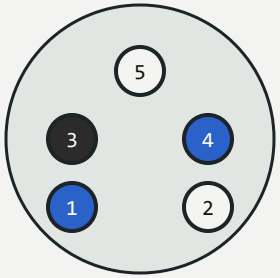
Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness WheelSpeed-Front

VRC front IN plug

`VRC-F-IN · LP-16-J05PE-02-001`

FACE SOURCED

VRC front IN plug



mating face view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	`cab_fl_p`	FL wheel speed +	ABS Wheel Speed FL · pin 1	`solder cup (LP-16, S16)`
2	20 White	`cab_fl_n`	FL wheel speed -	ABS Wheel Speed FL · pin 2	`solder cup (LP-16, S16)`
3	20 Black	`w_ws_f_shield`	Front sensor shields	Splice to: Shield, not connected at the device end	`solder cup (LP-16, S16)`
4	20 Blue	`cab_fr_p`	FR wheel speed +	ABS Wheel Speed FR · pin 1	`solder cup (LP-16, S16)`
5	20 White	`cab_fr_n`	FR wheel speed -	ABS Wheel Speed FR · pin 2	`solder cup (LP-16, S16)`

Pin numbers: CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19) · harness WheelSpeed-Front

ABS Wheel Speed FR

WSS-FR · DT06-2S-CE05 + W2S

FACE SOURCED

ABS Wheel Speed FR



latch top

wire side (where the contacts go in) view · source

PIN	WIRE	WIRE ID	WHAT IT DOES	GOES TO	CONTACT
1	20 Blue	cab_fr_p	FR wheel speed +	VRC front IN plug · pin 4	0462-201-16141
2	20 White	cab_fr_n	FR wheel speed -	VRC front IN plug · pin 5	0462-201-16141

Pin numbers: Deutsch DT / DTM cavity numbers (moulded on the housing) · harness WheelSpeed-Front